

WHO BENEFITS FROM SURGE PRICING?

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New technologies have recently led to a boom in real-time pricing. I study the most salient example, surge pricing in ride hailing. Using data from Uber, I develop an empirical model of spatial equilibrium to measure the welfare effects of surge pricing. The model is composed of demand, supply, and a matching technology. It allows for temporal and spatial heterogeneity as well as randomness in supply and demand. I find that, relative to a uniform pricing counterfactual in which Uber sets the overall price level, surge pricing increases total welfare by 2.15% of gross revenue. Welfare effects differ substantially across sides of the market: rider surplus increases by 3.57% of gross revenue, whereas driver surplus and the platform's current profits decrease by 0.98% and 0.50% of gross revenue, respectively. Riders at all income levels benefit. Among drivers, those who work long hours are hurt the most, especially women.

KEYWORDS: Surge pricing, dynamic pricing, ride hailing.

1. INTRODUCTION

UNTIL AROUND 2010, FEW COMPANIES ADJUSTED PRICES in real time in response to supply and demand, with some exceptions like airlines, hotels, and electricity operators. New technologies, however, have led to rapid changes. Companies can now use the internet and smartphones to communicate prices instantly, and they can use rich data sets to design better pricing algorithms. More and more companies now set prices in real time, especially in digital markets. This can be desirable in terms of efficiency, since more flexible prices allow markets to clear. However, real-time pricing often hurts some market participants, raising concerns about redistribution.

Ride-hailing platforms like Uber and Lyft have become the most salient adopters of real-time pricing—or surge pricing, as Uber calls it. Despite potential efficiency gains, surge pricing has been widely criticized. Some people suggest that it can hurt riders, calling it a form of price discrimination or price gouging (Dholakia (2015)). Others suggest that it can hurt drivers, whose earnings might drop too low unless they carefully plan their actions around surge pricing (Goncharova (2017)). In response to these concerns, cities like Honolulu, New Delhi, and Singapore have banned or capped surge pricing. Some ride-hailing companies have also voluntarily chosen to avoid surge pricing; DiDi (the largest platform in China) stopped using it, adopting instead potentially inefficient

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queues. However, there is limited evidence whether moving away from surge pricing actually benefits riders and drivers. This paper aims to fill that gap.

In this paper, I develop an empirical model of ride hailing to determine who gains and who loses from surge pricing. The model allows me to measure the welfare effects—on riders, drivers, and the platform—of moving from uniform pricing, where prices only depend on trip distance and duration, to surge pricing. The model is composed of three parts: demand, supply, and a matching technology. On the demand side, riders decide whether to open the app and whether to request a trip. On the supply side, drivers decide when to start and stop working and where to move when they are available. The matching technology captures matching and search frictions: it determines which rider is matched to which driver, and, thus, how long riders need to wait for pickup and how long drivers need to wait between trips. The model accounts for high-resolution spatial and temporal heterogeneity as well as transient random shocks. I integrate all pieces into a model of spatial equilibrium to simulate market behavior under alternative pricing policies.

My empirical analysis uses Uber data from Houston in March–April 2017, a period during which Uber was the only ride-hailing platform. I focus on a region in the center of Greater Houston that is 1.3 times the area of Washington, D.C. and where most of the surge pricing activity in the city takes place. I observe high-resolution data on drivers and riders. I also see detailed data on prices, which Uber sets as the base fare (a linear function of the trip distance and duration) times a *surge multiplier* that varies in real time and depends on the origin of the trip.

To estimate agents' short-run response to prices (how real-time price changes affect drivers' movement and exit decisions as well as riders' trip request decisions), I exploit rounding in the surge pricing algorithm, as in Cohen, Hahn, Hall, Levitt, and Metcalf (2016). I identify riders' response to pickup times—and, hence, the value of time—from variation that arises from drivers' exact position relative to riders. My data do not have exogenous variation from which to estimate long-run responses (how decisions to open the app depend on expected prices), so I calibrate them to experimental estimates. I use the supply elasticities that Caldwell and Oehlsen (2022) estimate for Houston, and I use demand elasticities that I estimate from experiments that Uber ran in Latin America.

I find that surge pricing increases total welfare by 2.15% of gross revenue—or \$0.25 per trip—relative to uniform pricing.¹ However, the effects on different sides of the market are asymmetric. Whereas rider surplus increases by 3.57% of gross revenue, driver surplus decreases by 0.98% of gross revenue. Quantifying the welfare effect on Uber is more challenging. Uber's current profits decrease by 0.50% of gross revenue. This probably does not mean that Uber is hurt, but rather that it is willing to forgo current profits to increase rider surplus—the right choice if it believes that long-run shareholder value is more closely tied to rider surplus and customer retention than to current profits. Consistent with this interpretation, I find that the overall price level in the data is well below the level that maximizes current profits but close to the point that maximizes rider surplus.

The asymmetry in welfare effects across riders and drivers can be decomposed into three parts, each of which arises from the way surge pricing interacts with matching inefficiencies. First, surge pricing saves people's time as it mitigates imbalances between supply and demand: riders are picked up more quickly, and drivers wait less time between trips. These time savings result in larger welfare gains for riders. My estimates imply that riders have a high value of time, which is as expected, given that they tend to travel during

¹Welfare effects are somewhat small because surge pricing only affects 21.2% of trips (i.e., the surge multiplier is above 1). For the remaining 78.8% of trips, the multiplier is equal to 1.

time-sensitive moments. The value of time to drivers, on the other hand, is their average hourly earnings, which are slightly above minimum wages.

Second, apart from time savings, surge pricing also increases rider surplus by allocating trips more efficiently. When drivers are scarce, surge pricing assigns trips to riders who have a high willingness to pay, rather than randomly (as uniform pricing does). But there is no such increase in driver surplus. Drivers observe little information about trips before accepting them—only the surge multiplier and the distance to the rider. Thus, apart from the time savings mentioned above, surge pricing can only play a limited role in eliciting drivers' preferences and allocating trips to drivers that value them most.

Third, prices are lower on average with surge pricing, decreasing driver surplus and the platform's current profits. I assume that, to maximize shareholder value, Uber maximizes a weighted sum of current profits, rider surplus, and driver surplus. I back out the weights that rationalize how Uber prices in the data. I find that when constrained to a uniform multiplier, the platform sets it above the average multiplier with surge pricing.² This is the case because, for a given time and place, it is worse to err by setting prices too low than too high, in part because of a matching failure—which [Castillo, Knoepfle, and Weyl \(2023\)](#) analyze theoretically—that hurts all market participants when drivers are scarce. Under uniform pricing, the only way to avoid that failure is with a high multiplier at all times. With surge pricing, the platform can set a lower average price, and surge pricing automatically increases prices during high demand times, avoiding the problem.

I also analyze heterogeneous effects among riders and among drivers. Surge pricing benefits riders at all income levels, but low-income riders benefit the most. They gain from lower prices, shorter pickup times, and more reliable trips. High-income riders also benefit, but they would prefer higher prices, which would further shorten pickup times and make trips more reliable. On drivers' side, I find that surge pricing especially hurts women and drivers who work many hours per week: they tend to work at times during which surge multipliers are low.

The public debate about the desirability of surge pricing has emphasized its negative effects on riders and drivers. My results suggest that riders' complaints are not well-founded. Their confusion might arise because they are unaware that they would pay higher average prices without surge pricing, and because they do not account for the indirect effects of eliminating surge pricing—longer pickup times and lower reliability. On the other hand, my findings suggest that drivers might have good reason to complain. Their hourly earnings are not much higher than the minimum wage, so even the small effects I find might be a concern.

My analysis has some limitations. I calibrate long-run responses to experimental estimates that are not ideal—because they come from different cities or because they measure a response that is not exactly the relevant one. I thus present robustness checks in which I vary the magnitude of long-run responses. Another limitation is that surge pricing could bring additional efficiency gains if drivers saw destinations before accepting trips (especially if multipliers varied by destination), but that is not the case in my data, so I cannot quantify this potential benefit.³

Other limitations relate to the external validity of my results. Uber was the only ride-hailing platform in Houston, and, thus, it provides a clean environment in which to analyze surge pricing in the absence of competition. Generalizing my results to markets

²This result hinges on the assumption that Uber maximizes the same weighted sum in a uniform pricing counterfactual and in the status quo.

³Uber recently started showing the destination to drivers in some places—in California starting in January 2020, and in another 24 cities in the U.S. starting in March 2022.

where multiple platforms compete presents some challenges. First, one needs to understand how riders and drivers substitute between platforms. An important aspect of this is the extent to which riders and drivers multi-home. For example, if drivers use multiple platforms simultaneously, available drivers become a common-pool resource shared between platforms, reducing the incentive to price high to offer a better service quality, as discussed by [Castillo, Knoepfle, and Weyl \(2023\)](#). Also, grasping how competition affects platforms' incentives is vital. Network effects may bring about fierce competition to become the dominant platform, leading to an even larger emphasis on rider surplus relative to current profits.

Another limitation is that Houston is a low-density city with poor public transit, so my findings are more likely to apply to similar cities, like Los Angeles or Miami, than to cities with a higher density or with better public transportation.⁴ Lastly, I focus on a relatively high-density central region in which most surge pricing activity occurs, so I cannot say much about how surge pricing affects low-density suburbs.

Related Work

Two earlier papers analyze the welfare effects of surge pricing in ride hailing. [Cachon, Daniels, and Lobel \(2017\)](#) propose a theoretical model, and [Ming, Tunca, Xu, and Zhu \(2023\)](#) estimate an empirical model of DiDi in Beijing. Both find that all participants benefit from surge pricing, although riders can be hurt at times. Improving upon these studies, I use detailed data to model and estimate matching frictions—the main driver of my results—which, in my view, explains why my findings differ from theirs. The theoretical work by [Castillo, Knoepfle, and Weyl \(2023\)](#) highlights this issue. They point out important matching inefficiencies that arise with excess demand, which can be avoided either with a high uniform price or with surge pricing and lower average prices—a solution that potentially hurts *drivers* (as I confirm empirically).

Many computer science and operations research papers have analyzed surge pricing ([Besbes, Castro, and Lobel \(2021\)](#), [Garg and Nazerzadeh \(2022\)](#)). A survey by [Yan, Zhu, Korolko, and Woodard \(2020\)](#) gives a detailed overview. The main goal of these studies is to improve the design of surge pricing algorithms. In contrast, I take the design of the algorithm as a given and analyze its effect on market participants.

Methodologically, my work relates to an empirical literature on matching and spatial equilibrium in urban transportation. Two works study taxi markets. [Fréchet, Lizzeri, and Salz \(2019\)](#) analyze entry restrictions and the adoption of Uber-like matching. [Buchholz \(2021\)](#) finds that changes to the structure of taxi fares can decrease search inefficiencies.⁵ Other papers study ride hailing. [Ghili and Kumar \(2021\)](#) focus on economies of density, and [Rosaia \(2023\)](#) focuses on competition between platforms. These papers do not have data on riders—only on drivers and trips—and so they rely on structural assumptions to back out demand. In contrast, I observe riders and whether they request trips, so I estimate demand directly from the data.

Two recent papers study dimensions of pricing in ride hailing other than surge pricing. [Buchholz, Doval, Kastl, Matjka, and Salz \(2022\)](#), who measure heterogeneity in the

⁴A city with better public transit would mainly differ from Houston in that demand is more elastic (and consumer surplus is smaller) because riders have better outside options. My main qualitative findings hold in such a scenario (Appendix E), but welfare effects are somewhat smaller due to more moderate changes in consumer surplus.

⁵[Brancaccio, Kalouptsi, Papageorgiou, and Rosaia \(2023\)](#) use a similar methodology to model the bulk shipping industry.

value of time, quantify welfare under different forms of price discrimination that exploit this heterogeneity. Their estimates of the average value of time are lower than mine, but they are also above median wages and well above minimum wages. [Gaineddenova \(2022\)](#) examines the desirability of centralized versus decentralized pricing.

Some papers estimate demand or supply in ride-hailing markets. [Cohen et al. \(2016\)](#) measure rider surplus. My identification strategy is closely related to theirs: we both exploit rounding in the surge pricing algorithm. On the supply side, [Angrist, Caldwell, and Hall \(2021\)](#) and [Chen, Chevalier, Rossi, and Oehlsen \(2019\)](#) estimate the degree to which drivers value flexibility. [Cook, Diamond, Hall, List, and Oyer \(2020\)](#) find a gender earnings gap and [Caldwell and Oehlsen \(2022\)](#) find differences in labor supply elasticities by gender. Relative to these works, I estimate both sides of the market and integrate them in an equilibrium model to analyze welfare effects.

Finally, my paper adds to the literature on platforms. Theoretical works assume simple forms for matching frictions ([Rochet and Tirole \(2003\)](#), [Weyl \(2010\)](#)), and conclude that welfare effects are determined by elasticities and externalities. In contrast, my findings are driven by details of the matching frictions that I microfound and estimate. This highlights that studying markets in which search and matching are central requires modeling frictions carefully ([Gavazza and Lizzeri \(2021\)](#)).

2. SETTING AND DATA

I analyze the Uber market in Houston between March 16 and April 8, 2017 ([Uber \(2024\)](#)). I focus on UberX, Uber's main product, which matches riders to independent drivers.⁶ Lyft did not operate in Houston between November 2014 and May 2017.⁷ Thus, Houston is a clean setting in which to analyze the welfare effects of surge pricing without having to consider competition between platforms.

I focus on trips that start in the region I show in Figure 1a, where most surge pricing activity in Greater Houston takes place. It accounts for 53.2% of all trips, but for 85.7% of trips with a surge multiplier above 1. Its area is 203 km² and its population is 520,000. This region covers the most dense areas of the city, including the two major business districts and a mix of residential areas ranging from upscale to low-income neighborhoods. Supplemental Appendix 2.1 ([Castillo \(2025\)](#)) presents further details about it and how it compares to the rest of Greater Houston. By focusing on this region, I do not account for the effects of surge pricing on suburbs; however, as in other cities, those areas have little surge pricing activity, so effects are likely to be small.

Riders, Drivers, and Trips

The raw data have a high temporal and spatial resolution. I observe the GPS location of riders and drivers—with anonymized identifiers—every few seconds whenever the app is open. I aggregate data at the level at which surge pricing varies: into two-minute *periods*, which I index by $t \in T$, and into the hexagons in Figure 1a, which I call *locations* and index by $l \in L$. There are 1681 hexagons in the region of analysis; each one is roughly 400 meters across. This level of aggregation is much finer than in previous papers, such as

⁶UberPool, which matches riders going in similar directions, was not available. Other products, such as UberXL (larger cars) and UberBlack (luxury cars), accounted for less than 5% of trips.

⁷Lyft decided to exit Houston after the City Council passed an ordinance requiring extensive background checks for drivers, including fingerprinting and a physical exam.

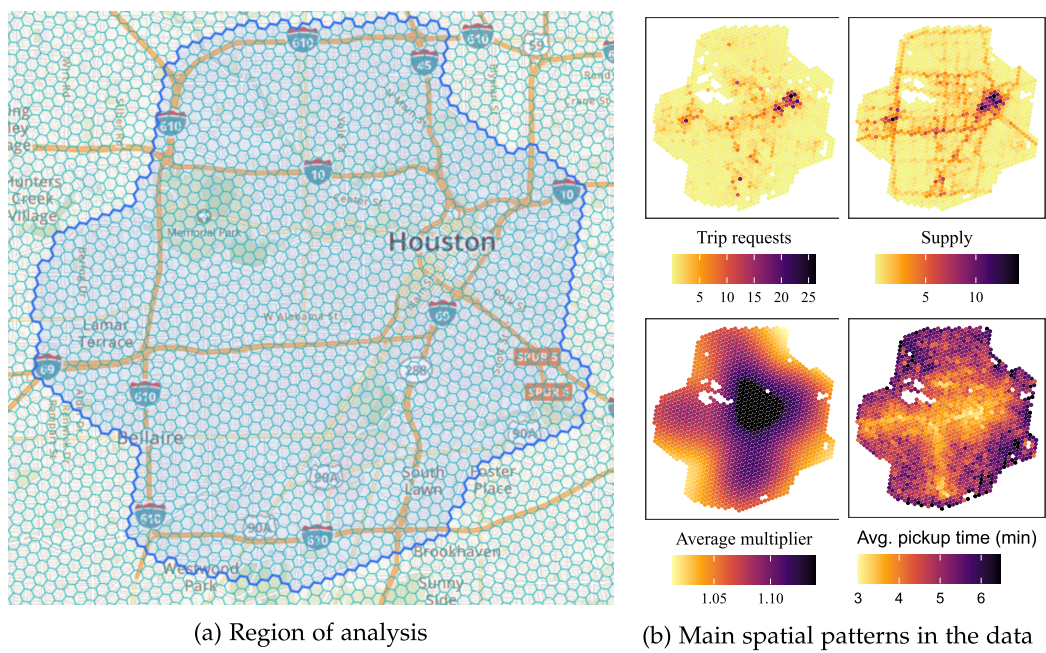


FIGURE 1.—Region of analysis and main spatial patterns. *Note:* Subfigure (a) shows a map of the center of Greater Houston. The shaded area is the region of analysis. Surge pricing varies spatially at the hexagon (or *location*) level. Subfigure (b) shows the main spatial patterns in the data. The upper panels show the number of trips that take place from every location and the average number of available drivers in every location (both are normalized to have mean 1). The bottom panels show the average surge multiplier and pickup time by location.

Buchholz (2021) and Fréchette, Lizzeri, and Salz (2019). A higher resolution is important to capture the short-run, local imbalances that surge pricing is meant to counteract.

When a rider selects a destination—in which case I say she is *active*—the app shows a fare to her destination and an estimated time to pickup (or *pickup time*, for short). I observe all variables: origin, destination, fare, and pickup time. I aggregate rider data by *session*, defined as a period of activity with no gaps of half an hour or longer. Sessions can end in two ways: the rider can request a trip, or she can be inactive for half an hour, after which I say she *leaves* the app.⁸ I take the destination, fare, and pickup time to be the last ones the rider observed during the session.

I use the trip history to identify riders’ home location, which I match to median income by census tract (Supplemental Appendix 2.2). I call this variable *income*, with the understanding that it is a proxy. I identify with confidence the home of riders that account for 77.3% of trips. These riders tend to take many trips and have a high conversion rate, so I call them *frequent riders*. The remaining trips were taken by *occasional riders*, who took too few trips to allow me to determine where they live.⁹

I also observe detailed data on drivers. When they are online, I see their location and their state: they can be available to be matched (45.2% of the time), picking up a rider

⁸If the rider requests a trip but cancels it before pickup, I assume the rider never requested it.
⁹On average, frequent riders took 4.73 trips in my sample and their conversion rate is 81.2%. Occasional riders took 1.67 trips on average, and their conversion rate is 52.7%.

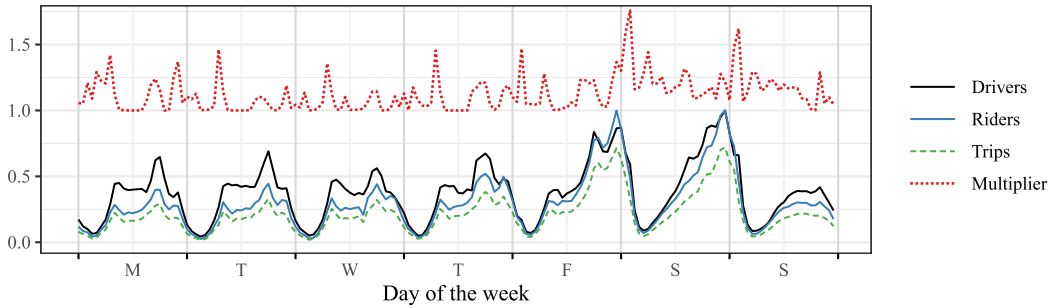


FIGURE 2.—Weekly patterns in demand, supply, trips, and surge multipliers. *Note:* These time series represent the average number of drivers working, the average number of riders that open the app (i.e., the number of rider sessions), the average number of trips that take place, and the average surge multiplier (weighted by rider session) over the course of a week. Riders and drivers are normalized to have a maximum of 1. Trips are on the same scale as riders.

(17% of the time), or taking a rider to the destination (37.8% of the time).¹⁰ I also observe drivers' gender and *experience* (how many trips they have completed). Twenty-two point one percent of drivers are women, and they account for 15.6% of the total hours worked.

Based on drivers' experience and work schedules, I classify them into seven groups using a *k*-means clustering method (Supplemental Appendix 2.3). A clearly distinct cluster, which accounts for 13.4% of drivers and 36.8% of the total hours worked, is composed of highly experienced drivers who work many hours per week, mostly during the day and evening. I call them *full-time* drivers. The other clusters are composed of less experienced drivers who work fewer hours; the main difference between these clusters is that drivers tend to work during quite distinct times of the week. I refer to drivers in these six clusters as *part-time* drivers.

Finally, I observe trip statistics, including the time and location during request, pickup, and dropoff, as well as the trip price and Uber's commission. Figure 2 plots market behavior over the week. Demand and supply behave similarly, with two noticeable differences: there is excess supply around noon during weekdays and excess demand during Friday and Saturday evenings. The upper panels in Figure 1b show spatial patterns. The area with most trip requests, towards the northeast, is Downtown. Another high demand area is Uptown, on the west. Drivers tend to work in areas that have a large number of trip requests and along highways.

Trip Request, Matching, Fares, and Surge Pricing

The trip request process starts with the rider selecting a destination.¹¹ The app then shows her a fare in dollars and a pickup time.¹² If the rider requests a trip, it is offered to the nearest available driver, who has a few seconds to accept. If the driver does not accept, the trip is offered to the second nearest available driver, and so on. The driver does not know the trip destination before the rider gets in the car. Once the trip is completed, the

¹⁰Drivers can open the app without being available for matches, which I count as being offline.

¹¹This was the process as of 2017. Following subsequent changes, Uber now groups trips into batches that are matched every few seconds, surge pricing no longer uses rounding, and surge pricing is no longer multiplicative but additive for drivers (Garg and Nazerzadeh (2022)).

¹²This differs from the way the market operated until 2016. Back then, the rider did not have to choose a destination to request a trip and was shown a surge multiplier instead of a fare in dollars.

rider pays the fare by credit card. Uber takes a booking fee of \$2.30 per trip. The rest of the fare is split between the driver and Uber, which takes a commission that varies between 24% and 28%, depending on how long ago the driver joined Uber.

The fare is the product of two components. The first is the *base* or *unsurged fare*, which is equal to $\$3.30 + \$0.87 \times (\text{predicted miles}) + \$0.11 \times (\text{predicted minutes})$. Uber predicts the distance and duration of the trip based on the pickup and dropoff coordinates and the hour of the week. Importantly, the base fare does not change in real time. Given an origin and destination, it only varies to the extent that the predicted distance and duration change—because of expected traffic congestion, for instance—but it does not vary in response to supply or demand.

The second component of the fare is a *surge multiplier* that responds in real time to supply and demand. Spatially, surge multipliers vary by the locations shown in Figure 1a. They are simultaneously updated every two minutes across the whole city. Whenever a driver is available, he can see a map of all multipliers.

Figure 2 shows that surge multipliers tend to be high when demand is high relative to supply: in the early morning, during the morning and afternoon rush hour, and during the weekend—especially during bar hours. The bottom two panels in Figure 1b show maps of the average multipliers and pickup times. Multipliers are highest around Downtown and towards the south. Pickup times tend to be lowest where most available drivers are located. Supplemental Appendix 5.1 shows the distribution of surge multipliers weighted by rider session. The average multiplier is 1.149. For 78.8% of riders, the underlying multiplier of the fare they see is 1. Less than a third of the variation in surge multipliers is predictable: a regression of the multiplier on half-hour-of-the-week-by-location fixed effects has an R^2 of 0.284.

3. MODEL

I start with an overview of the model of a ride-hailing market. Agents make two types of decisions: long-run and short-run. In the long run, they decide whether to enter the market. Riders choose whether to open the app given their expectations about prices and pickup times. Drivers decide whether to start working depending on how much they expect to earn. These decisions are based on expectations because agents must make adjustments ahead of time. A driver might have to arrange for someone to take care of his children, for instance, and a rider might open the app unless she borrows her spouse's car.

In the short run, agents who already entered the market make choices using the information they observe in the app. Riders observe a fare and a pickup time, based on which they decide whether to request a trip. Available drivers observe a map of surge multipliers, based on which they decide whether to keep on working and where to move.

The platform also takes some actions. It computes surge multipliers, fares, and pickup times, and it matches riders and drivers.

Figure 3 is a timeline that shows everything that happens during one two-minute period. It clarifies the mechanics of the interaction between agents.

3.1. Demand

Riders make two decisions. First, they decide whether to open the app and choose a destination. Second, after observing the fare and pickup time, they decide whether to request a trip.

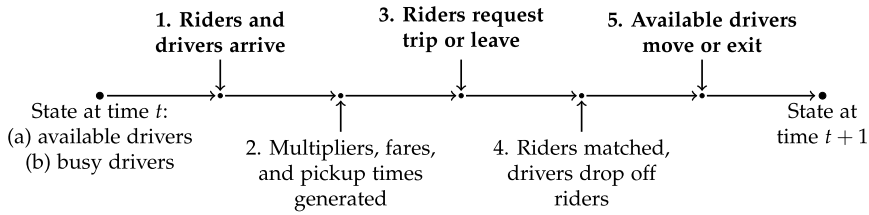


FIGURE 3.—Model timeline. *Note:* Timeline of the events that take place during every two-minute period. The initial state is the set of available and busy drivers. Each available driver is in a certain location, and every busy driver will drop off a rider and become available during some future period in a certain location. Steps during which riders and drivers make decisions are highlighted in bold above the timeline. Steps below the timeline are mechanical actions taken by the platform.

I model riders as static decision makers. In reality, they can decide to wait and request a trip later. However, waiting for a lower multiplier is not an important response to surge pricing in the data.¹³ Some riders could also be worried about the price of the return trip, but this is a concern for at most one out of four riders.¹⁴

I now explain how riders make their choices in reverse order.

3.1.1. Trip Request

Rider i opened the app in location l during hour-of-the-week h and chose destination k . She is characterized by a vector of covariates x_i . The Uber app shows her a price p_i and a pickup time w_i (w for “waiting”). If she requests a trip, her utility is

$$U_i = \alpha(x_i, l, h) + \beta(x_i)p_i + \gamma(x_i)w_i + \epsilon_i. \quad (1)$$

The first term captures patterns in the value of a trip by location, hour of the week, and covariates. The second term captures the disutility from paying, and the third term captures the disutility from waiting.¹⁵ Finally, ϵ_i is an error term. Utility is measured relative to a short-run outside option the rider chooses if she does not request a trip. It may be an alternative form of transportation—driving, biking, or taking a bus—or simply not going to her destination. She requests a trip if $U_i > 0$.

The key parameters are the price and pickup time coefficients, $\beta(x_i)$ and $\gamma(x_i)$, which measure price and pickup time elasticities. Both depend on covariates. The ratio $\frac{\gamma(x_i)}{\beta(x_i)}$ measures the *value of time*, the value of spending time at the destination minus the value of spending time at the origin (as in Buchholz et al. (2022)).

3.1.2. Opening the App

In addition to their short-run outside option, riders have a long-run outside option that requires some planning; thus, they can only choose it before they observe prices and pickup times. For instance, riders might buy a car or coordinate to carpool if they expect

¹³As the multiplier goes from 1 to 1.5, the fraction of riders who request a trip or leave the platform in the first two minutes only decreases from 73% to 68%, and to 66% if the multiplier is above 1.5.

¹⁴Only 24.2% of trips start within 1 km of the previous dropoff location, and only 15.8% of trips do so if I only count cases in which the previous trip started at the rider’s home.

¹⁵Appendix E shows that the welfare results change little if, rather than responding to the shown pickup time, riders respond to the expected pickup time they can infer from the former.

high prices. Let u_i be the value of this outside option for rider i , relative to her short-run outside option. It is drawn from a distribution F^u .

Let $U_{lhx_i} = \mathbb{E}[\max\{\frac{1}{\beta(x_i)}U_i, 0\} | l, h, x_i]$ be rider i 's ex ante dollar value of opening the app. It is her expectation of the value of her best choice—either requesting a trip, with value $\frac{U_i}{\beta(x_i)}$, or the short-run outside option, with value 0—given what she knows *before* opening the app and observing the price and the pickup time. This is an equilibrium quantity: it depends on the distribution of prices and pickup times that rider i faces in equilibrium. She opens the app if $U_{lhx_i} > u_i$.

There is an arrival rate λ_{lhx}^0 of riders with covariates x that might enter the market in location l during hour-of-the-week h . Each potential entrant i opens the app and chooses a destination with the probability that $U_{lhx_i} > u_i$, so the actual rate at which riders enter is $\lambda_{lhx} = \lambda_{lhx}^0 F^u(U_{lhx})$. I assume that $F^u(u) \propto u^\rho$.¹⁶ Thus,

$$\lambda_{lhx} = A_{lhx} U_{lhx}^\rho \quad (2)$$

for some demand shifter A_{lhx} . Rider surplus, relative to the long-run outside option, is $RS_{lhx} = \int_0^{U_{lhx}} (U_{lhx} - u) dF^u(u) = \frac{A_{lhx}}{1+\rho} U_{lhx}^{\rho+1}$.

The key parameter of this arrival model is ρ . It determines the long-run elasticity of demand—how the number of requests changes if there is a change in prices or pickup times that riders know of in advance.¹⁷

3.2. Supply

Drivers make three decisions. At the beginning of a shift, they decide whether to start working. If they do, every time they are available they decide whether to keep on working or exit the platform. Finally, every time they stay, they choose where to move before the beginning of the next period. Sections 3.2.1–3.2.3 describe these three decisions in detail, working backwards from movement to entry decisions.

I now describe some general features that are relevant for all three decisions. Driver j belongs to a discrete type d , which captures the driver's gender and preferred shift patterns. His utility takes the form

$$V_j = \sum_t (\pi_{jt} - k_{tj}), \quad (3)$$

where the sum is over all periods that he worked. The term π_{jt} denotes net earnings—earnings minus the physical cost of driving (fuel, depreciation, and maintenance)—and k_{tj} represents the per-period opportunity cost of working.¹⁸ With this utility function, two drivers with an equal opportunity cost who get the same net earnings are equally well off. It does not matter where in the city they drove, whether they drove in traffic, or whether they were busy or idle most of the time. As will become clear, this is a simplifying assumption that I need for tractability.

¹⁶This distribution is unbounded, but it can be interpreted as the left tail of the actual distribution.

¹⁷The arrival decision depends on expected prices and pickup times through U_{lhx_i} . This implicitly assumes that the value of time is the same in the short and in the long run.

¹⁸If k_{jt} increases weakly over time (an assumption that I impose in Section 3.2.2), the disutility of working is concave in the time worked, as in standard labor supply models. This functional form ignores income effects. This is unlikely to be problematic unless drivers depend on Uber for a large fraction of lifetime income and expect pricing changes to be permanent.

3.2.1. Movement

Driver j of type d is available in location l at time t during hour-of-the-week h , and he observes surge multipliers $\mathbf{m}_t$. He already decided to keep on working. The state—the information observed by the driver—is $\mathbf{s}_t = (l, h, \mathbf{m}_t)$.

I assume that the driver's location in period $t + 1$ follows a stochastic movement rule that depends on two things. First, it depends on mean future earnings given the state: the driver is more likely to move to locations where, on average, earnings are higher. Mean earnings do not depend on private information the driver has about what he might do next. Instead, they are simply an empirical average of the earnings drivers obtain for the next $\bar{t}$ periods if they move to some location k after being in state $\mathbf{s}_t$. Second, the movement rule follows road and traffic patterns. The driver cannot move instantaneously to the other end of the city, and he is less likely to move to far-away locations during rush hour than he is at 4 am.

Let $v_{dk}(\mathbf{s}_t)$ be *mean future earnings*, the mean of the net earnings type- d drivers get between periods $t + 1$ and $t + \bar{t}$ if they move to location k after being in state $\mathbf{s}_t$,

$$v_{dk}(\mathbf{s}_t) = \mathbb{E} \left[\sum_{s=t+1}^{t+\bar{t}} \pi_s \middle| d, l_{t+1} = k, \mathbf{s}_t \right] - c_l^k, \quad (4)$$

where π_t denotes net earnings during two-minute period t . I denote the location at time t by l_t , and c_l^k is the physical cost of moving from location l to k . I set $\bar{t} = 45$ because, in the data, most of the effect of surge multipliers on earnings takes place during the first 90 minutes. Thus, $v_{dk}(\mathbf{s}_t)$ incorporates almost all information about future earnings drivers can infer from multipliers.

Mean future earnings $v_{dk}(\mathbf{s}_t)$ are an empirical average of market behavior in equilibrium. There are many possible future outcomes for type- d drivers who move to k after being in state $\mathbf{s}_t$. They might subsequently move north or south, multipliers might go up or down, and they might have to wait or be matched immediately. The probability of these outcomes depends on equilibrium behavior—how drivers behave after moving to k , how every other driver in the market behaves, and how riders behave. The mean of net earnings over all these possibilities is $v_{dk}(\mathbf{s}_t)$.

Let $l_{j,t+1}$ be the location to which driver j moves. The movement rule is

$$\Pr(l_{j,t+1} = k | \mathbf{s}_t) = \frac{\exp(\overbrace{\omega_{lkhd}}^{\text{Roads/traffic}} + \overbrace{\delta_d v_{dk}(\mathbf{s}_t)}^{\text{Earnings}} + \overbrace{\zeta_{kt}}^{\text{Unobserved shock}})}{\sum_{k'} \exp(\omega_{lk'hd} + \delta_d v_{dk'}(\mathbf{s}_t) + \zeta_{k't})}. \quad (5)$$

The first term inside the exponential are fixed effects that capture road and traffic patterns. If locations l and k are far from each other or have no roads connecting them, then ω_{lkhd} is very low, so the probability of moving to k in one period is negligible. If it is easy to get from l to k in two minutes at 4 am but not in traffic, then ω_{lkhd} is higher at 4 am than during rush hour. ω_{lkhd} also varies by driver type to capture the tendency of some types of drivers to work in some areas of the city at some times of the week.

The middle term captures the fact that drivers are more likely to go to locations that have higher mean future earnings. The key parameter in this model is δ_d , which measures the extent to which drivers are more likely to move towards areas with high earnings. Drivers do not respond to surge multipliers directly, but they respond indirectly: higher

multipliers lead to higher expected earnings, so δ_d also measures how movements respond to multipliers. Finally, ζ_{kt} is an unobserved term that captures systematic shocks that cause drivers to flock towards specific locations, as might happen if drivers expect an event to end soon.

Drivers are not fully rational—that is, forward-looking and utility-maximizing—as they are in Buchholz (2021) and Fréchet, Lizzeri, and Salz (2019). It is not feasible to model fully rational drivers in this setting because of the curse of dimensionality: the state involves all multipliers surrounding the driver. The movement rule (5), however, is a tractable model of driver movement that captures the essence of rational behavior. Drivers are more likely to move towards high-earnings areas in a forward-looking manner: by accounting for mean future earnings, drivers implicitly consider where they will move in subsequent periods, whether surge multipliers might go up, and whether they might get a trip quickly. Moreover, with this model, I can avoid the curse of dimensionality by assuming that agents form beliefs $v_{dk}(s_t)$ as a lower-dimensional regression of earnings on the state, as I explain in Section 3.5.

This model departs from a standard dynamic model of a rational agent in three ways.¹⁹ First, drivers do not respond to earnings more than $\bar{t}$ periods into the future. Second, drivers respond to earnings until period $t + \bar{t}$, even if they plan to stop working before then. Third, $v_{dk}(s_t)$ does not account for future fixed effects ω_{lkhd} , unobserved terms ζ_{tk} , or errors underlying the movement rule—consistent with the assumption that utility is equal to net earnings minus opportunity costs. My model, thus, views the fixed effects and error terms as capturing constraints imposed by roads or traffic as well as mistakes from inattention or limited knowledge.

3.2.2. Exit

Moving backwards, I now describe drivers' decision whether to keep on working or exit. Driver j of type d is available in location l at time t during hour-of-the-week h . He stays if the expected value of staying is positive.

A key input for this decision is the per-period opportunity cost of working. I assume it takes the form $k_{jt} = \kappa_d \max\{t - \bar{t}_j, 0\} + s_{jt}$. The first term is zero until some time $\bar{t}_j$ and then it increases linearly over time at a rate κ_d , accounting for the fact that drivers get tired or eventually need to get back home. The time $\bar{t}_j$ when the opportunity cost starts increasing is equal to $\underline{t}_j + T_j$, the period $\underline{t}_j$ when the driver started working plus a driver-specific duration T_j that captures preferences for a short or a long shift. The term s_{jt} is an unobserved error. I assume for now that it is mean-zero and i.i.d., but I relax this assumption in Section 5.

The driver decides whether to stay or exit based on the state $\tilde{s}_{jt} = (\tau_{jt}, l, h, \mathbf{m}_t)$, where $\tau_{jt} = t - \bar{t}_j$ captures his preferences (at time t) for working longer or ending the shift soon. The driver computes $v_{d,\Delta}^S(\tilde{s}_{jt})$, the expected value of staying at a future period $t + \Delta$ conditional on state $\tilde{s}_{jt}$, using the following recursion:

$$v_{d,\Delta}^S(\tilde{s}_{jt}) = \underbrace{q_{\Delta}(\tilde{s}_{jt})}_{\text{Match probability}} \underbrace{v_{d,\Delta}^M(\tilde{s}_{jt})}_{\text{Utility until end of trip}} + (1 - q_{\Delta}(\tilde{s}_{jt})) \underbrace{V_{d,\Delta+1}^S(\tilde{s}_{jt})}_{\text{Cont. value if unmatched}} - \underbrace{\kappa_d \max\{\tau_{jt} + \Delta, 0\}}_{\text{Expected opportunity cost}}. \quad (6)$$

¹⁹In most dynamic discrete choice models, the assumption of rationality is needed to infer utility from choices. I can depart from it because of an unusual feature: I observe an objective measure of utility—net earnings—which I use to estimate continuation values $v_{dk}(\cdot)$ directly from the data.

The driver is matched with probability $q_\Delta(\tilde{s}_{jt})$, in which case he gets the utility until the end of the trip $v_{d,\Delta}^M(\tilde{s}_{jt})$. If he is not matched, the continuation value is $V_{d,\Delta+1}^S(\tilde{s}_{jt}) = \mathbb{E}_s[\max\{0, v_{d,\Delta+1}^S(\tilde{s}_{jt}) - s\}]$, the expected value if the driver chooses optimally in the next period. Lastly, regardless of whether he is matched or not, the driver incurs the expected opportunity cost $\kappa_d \max\{\tau_{jt} + \Delta, 0\}$.

At time t , the driver uses recursion (6) to compute the current value of staying $v_0^S(\tilde{s}_{jt}) - s_{jt}$. He stays if it is positive and exits otherwise.

This recursion differs from a standard value function in two ways, reflecting two simplifying assumptions. First, if he is matched, the driver accounts for the utility until the end of the next trip, but he does not account for the continuation value after the trip is over. In principle, this might be a problem: the driver could make the wrong decision if staying only one more trip has a negative expected utility, but that trip will take the driver to a profitable neighborhood so it is worth staying two or more trips. That, however, is unlikely because opportunity costs are increasing. Second, the continuation value assumes that, at time $t + \Delta + 1$, the driver will decide whether to stay based on his current information $\tilde{s}_{jt}$, without accounting for the fact that he will observe the new state $\tilde{s}_{j,t+\Delta+1}$ before making that decision. Without this assumption, drivers would need to know—and I would need to estimate—the state transition probabilities, which are very high-dimensional.

The key parameter in this exit model is κ_d . It determines to what extent drivers respond to high multipliers by staying longer. If κ_d is low, opportunity costs are flat and drivers are likely to extend their shift when they observe high multipliers. If κ_d is high, the extra earnings they get if they stay longer are small relative to the quickly increasing opportunity cost, so they are unlikely to extend their shift.

Exit decisions depend on drivers' beliefs about the value of staying $v_{d,0}^S(\tilde{s}_{jt})$, which are an equilibrium object. Similarly to the mean future value from Section 3.2.1, they are empirical averages of equilibrium market behavior. They weight all possible histories (being matched soon, having to wait, getting a trip, etc.) based on how likely they are in equilibrium, conditional on $\tilde{s}_t$. The state space is large; I avoid the curse of dimensionality by assuming that agents form beliefs as a lower-dimensional regression of market observables on the state (see Section 3.5).

3.2.3. Entry

Suppose driver j , whose shift duration preferences are characterized by duration T_j , might start working in location l at time t . He has an outside option that requires some planning—he might work at a full-time job or meet with friends—so he forgoes that option the minute he starts his shift. Let $\bar{V}_j$ be the value that he gives up if he starts working. It is drawn from a distribution F_d^V . The driver expects his utility to be $V_{lhdT_j} = \mathbb{E}[V_j|l, h, d, T_j]$ if he decides to work. This is an equilibrium quantity: it depends on how the driver will behave, and on how he expects other agents to behave. He starts working if $V_{lhdT_j} \geq \bar{V}_j$.

There is a rate μ_{lhdT}^0 of potential entrants of type d with duration T in location l during hour-of-the-week h . Each potential entrant j starts working with the probability that $V_{lhdT_j} \geq \bar{V}_j$, so the actual rate at which drivers enter is $\mu_{lhdT}^0 F^V(V_{lh})$. I assume that $F^V(V_{lhdT}) \propto V_{lhdT}^{\sigma_d}$,²⁰ so the entry rate is

$$\mu_{lhdT} = B_{lhdT} V_{lhdT}^{\sigma_d}, \quad (7)$$

²⁰This unbounded distribution can be understood to be the left tail of a very large distribution.

where B_{lhdT} is a horizontal demand shifter and σ_d is the elasticity of entry with respect to expected utility. Driver surplus for (l, h, d, T) , relative to the outside option of not working, is $DS_{lhdT} = B_{lhdT} \int_0^{V_{lhdT}} (V_{lhdT} - V') dF^V(V') = \frac{B_{lhdT}}{1+\sigma_d} V_{lhdT}^{\sigma_d+1}$.

The key parameter of this entry model is σ_d . It is a measure of the long-run elasticity of supply: it determines the extent to which the number of drivers who start working responds to changes in expected utility (and earnings).

This model does not allow drivers to start working in response to unexpectedly high multipliers. I find evidence that that margin is unlikely to be important: Multipliers can only predict 15% of unexpected variation in future multipliers more than ten minutes into the future. Therefore, unexpectedly high multipliers convey little information about the total earnings during a shift.^{21,22}

3.3. Uber's Decisions

Matching

When a rider requests a trip, the matching process determines which driver picks her up as well as how long it takes the driver to pick her up. Thus, it determines the magnitude of the matching inefficiencies.

In period t , let I_t be the set of riders that request a trip and let J_t be the set of drivers that can be offered a trip—those who are available as well as those who will drop off a rider during the next four minutes. Matches take place as follows. First, the platform computes a pickup time w_{ij}^p for every pair of a rider $i \in I_t$ and a driver $j \in J_t$. The pickup time is drawn from a distribution $G(\cdot | l_i, l_j, b_j, h)$ that depends on the rider's location l_i , the driver's location l_j , whether the driver is busy b_j (i.e., dropping off a rider), and the hour-of-the-week h .

Riders in I_t are matched sequentially in a random order. For every rider, her trip is offered first to the driver in J_t with the lowest pickup time, who accepts it with probability ϕ^a if he is available and probability ϕ^b if he is busy.²³ If he does not accept, the trip is offered to the next closest driver, who also accepts it with probabilities ϕ^a and ϕ^b , depending on whether he is available or busy. The process goes on until the rider is matched or until no available drivers remain within 10 km, in which case the rider does not get a trip and takes her outside option.

A distribution $G(\cdot | l_i, l_j, b_j, h)$ with higher pickup times means a larger matching inefficiency. Lower acceptance probabilities ϕ^a and ϕ^b result in matches with drivers who are farther away, and, therefore, a larger matching inefficiency.

Surge Multipliers

The platform generates surge multipliers using the exact algorithm Uber used during the period of my data. I cannot disclose details of the algorithm because it is proprietary, but, most importantly, it depends on the number of available drivers and on the number of riders who open the app. Both quantities are aggregated over nearby locations and over the last few minutes.

²¹This is not a problem for drivers who might start working in high-demand areas, but they account for very few drivers. Ninety-four percent of drivers start working outside the area of analysis and drive in.

²²Since 2020, Uber has sometimes sent notifications to some offline drivers when surge pricing is high to induce them to start working. These notifications were not used during my sample period.

²³In the data, acceptance rates are high—84.7%, on average—and they do not increase with surge multipliers (see Supplemental Appendix 4.1). This is so because Uber punished drivers who had low acceptance rates; the main reason drivers did not accept the remaining 15.3% of offers was inattention.

3.4. Other Parts of the Model

Appendix A presents three additional parts of the model that are mostly mechanical. First, I explain how Uber computes the pickup duration for every rider-driver pair. Second, I present a model for the trip destination and trip duration. It is based on empirical distributions and assumes a fixed distribution of the trip destination conditional on origin, hour of the week, and rider covariates x_i . Third, I explain how I account for the behavior of drivers when they are outside the region of analysis. I assume that their movements when they are far from that region are unchanged in counterfactuals.

3.5. Equilibrium

Sections 3.1–3.4 characterize the behavior of all agents, and the timeline in Figure 3 specifies how they interact. The only elements that remain to be specified for a complete model are agents' beliefs, which are determined by an equilibrium condition: beliefs must be consistent with empirical averages.

Let $\mathbf{U}$ be the vector of riders' ex ante utilities U_{lhx} , and let $\mathbf{V}$ be the vector of drivers' expected utilities V_{lhdT} . Let $\mathbf{v}$ and $\mathbf{v}^S$ be, respectively, the vectors of drivers' mean future earnings $v_{dk}(\mathbf{s}_t)$ and the expected value of staying $v_{d,0}^S(\tilde{\mathbf{s}}_{jt})$ for all possible states $\mathbf{s}_t$ and $\tilde{\mathbf{s}}_{jt}$. I define $\mathcal{X}$ as the set of all possible beliefs, so that every tuple of beliefs $\mathbf{x} = (\mathbf{U}, \mathbf{V}, \mathbf{v}, \mathbf{v}^S)$ belongs to $\mathcal{X}$.

Suppose the market behaves as described by the model under a pricing policy P . Let $f^P(\cdot) : \mathcal{X} \rightarrow \mathcal{X}$ be the function that maps a vector of beliefs $\mathbf{x}$ into the vector in $\mathcal{X}$ of empirical averages of the behavior of the market when agents' beliefs are $\mathbf{x}$.

A market equilibrium for pricing policy P is characterized by a vector of beliefs $\mathbf{x}^P \in \mathcal{X}$ that is a fixed point of $f^P(\cdot)$:

$$\mathbf{x}^P = f^P(\mathbf{x}^P). \quad (8)$$

This equation says that equilibrium beliefs are consistent with empirical averages. Appendix A proves that an equilibrium exists and that under an additional condition (which is always satisfied in simulations), the equilibrium is unique and stable.

3.5.1. Equilibrium Computation

To compute the equilibrium, I approximate $f^P(\mathbf{x})$ by simulation. Assuming agents' beliefs are $\mathbf{x}$, I simulate the history of the whole market during several weeks, agent by agent: I keep track of all surge multipliers, of every driver (where he was in every period, all trips he completed, and when he left), and of every rider and whether she requested a trip. I then use the simulated history to compute an empirical estimate $\hat{f}^P(\mathbf{x})$ whose expectation is $f^P(\mathbf{x})$.

Computing $\hat{f}^P(\mathbf{x})$. I compute empirical estimates of U_{lhx} and V_{lhdT} as averages of riders' and drivers' utilities in the simulated history. To estimate $v_{dk}(\mathbf{s}_t)$ and $v_{d,0}^S(\tilde{\mathbf{s}}_{jt})$, I cannot use simple averages because states are very high-dimensional, and so I run regressions that reduce the dimensionality of the problem.

Let Π_{jt} be driver j 's net hourly earnings from time $t + 1$ until $t + \bar{t}$ or the time he leaves, whichever comes earlier.²⁴ The empirical estimate of mean future earnings $v_{dk}(\mathbf{s}_t)$ that

²⁴To compute net earnings, I assume driving costs of \$0.26 per mile, the internal Uber estimate for the average UberX car in Houston, including fuel, maintenance, repairs, and depreciation.

I use is the prediction from the following regression, which I run on the data set of the simulated history:

$$\Pi_{jt} = \alpha(k, h) + h(\mathbf{m}_{tk}) + \chi_{jt}, \quad (9)$$

where k is the direction driver j moved to and $\alpha(k, h)$ is a flexible function that plays the role of location-by-hour-of-the-week fixed effects. The term $h(\mathbf{m}_{tk})$ is a flexible function of $\mathbf{m}_{tk}$, the vector of multipliers surrounding location k . I use a relatively simple smooth function that is radially symmetric. Supplemental Appendix 3.4 gives further details about this model.

The value of staying $v_{d,0}^S(\tilde{\mathbf{s}}_{jt})$ is not directly observable. To estimate it, I leverage a result that stems from the fact that the opportunity cost increases linearly over time. The value of staying can be expressed as a function of τ_{jt} (which captures whether the driver would like to work longer or end the shift soon) and of the expectation, conditional on $\mathbf{s}_t = (l, h, \mathbf{m}_t)$, of four observables: (1) earnings until the end of the next trip; (2) the duration until the end of the next trip; (3) the duration squared until the end of the next trip; and (4) match probabilities. Appendix C explains how one can compute the value of staying from these observables.

To obtain an empirical estimate of the value of staying, I estimate the four observables by running regressions of the form (9) on the data set from the simulated history, with the relevant observable on the left-hand side. I then use those estimates to compute the value of staying, as detailed in Appendix C.

Iterating to the Equilibrium. I compute the market equilibrium $\mathbf{x}^P$ by iterating on

$$\mathbf{x}_{n+1} = (1 - \gamma_n)\mathbf{x}_n + \gamma_n \hat{f}^P(\mathbf{x}_n), \quad \gamma_n \propto n^{-b}, b \in (0, 1), \quad (10)$$

which computes new beliefs as a convex combination of old beliefs and new empirical averages. Appendix A proves that this process converges to the equilibrium.

After finding equilibrium beliefs $\mathbf{x}^P$, I compute every other equilibrium quantity—for example, the decisions and surplus of agents, or the distribution of multipliers—by simulating many histories with beliefs $\mathbf{x}^P$ and computing averages.

3.5.2. Market Clearing

The market clears through a combination of prices (i.e., surge pricing) and waiting times. This is the case for both transient and steady-state imbalances between supply and demand.

Consider first short-run, transient imbalances. Suppose that one day, by pure chance (i.e., because of realizations of error terms), too many riders request trips from some neighborhood and deplete it of available drivers. Prices and pickup times go up, so fewer riders request trips. Drivers infer from high multipliers that earnings in that neighborhood will be high—trips will be more expensive, and drivers are scarce so they will be matched quickly—so they move in to the area and work longer. Riders' and drivers' responses, thus, rebalance the market.

Now consider long-run, steady-state imbalances. If there were, on average, a scarcity of drivers downtown during rush hour, surge multipliers and pickup times would both be high on average and drivers would spend little time waiting between trips. Facing a low expected utility, fewer riders would arrive, and those who arrive would request fewer trips. Facing a high expected utility, more drivers would start working, move downtown, and work longer if they are downtown. All these forces counteract the scarcity of drivers.

The equilibrium condition ensures that these margins adjust until beliefs are consistent with averages.

4. DESCRIPTIVE EVIDENCE AND IDENTIFICATION

I now show the variation in the data that I use to identify the main parameters of the model, and I explain informally my empirical strategy (Section 5 describes it formally).

4.1. Short-Run Demand Response

In the trip request model, the main parameters to identify are $\beta(x_i)$ and $\gamma(x_i)$, which determine how prices and pickup times affect the probability that a rider requests a trip. The main challenge is that prices and pickup times are endogenous.

Response to Prices

I estimate riders' response to price changes by exploiting a feature of the surge pricing algorithm, whose main steps are:

1. After analyzing supply and demand in the whole city, the algorithm computes a continuous *recommended multiplier* $\tilde{m}_{it}$ for each location.
2. Recommended multipliers are rounded to the nearest tenth (or to 1 if <1.15).
3. Rounded multipliers are smoothed out in space and time.
4. Smoothed multipliers are rounded to the nearest tenth (or to 1 if <1.15). The outcome m_{it} is the *surge multiplier*.

The final price shown to riders is $p_i = b + m_{it}(\bar{p}_i - b)$, where $\bar{p}_i$ is the *unsurged fare*, the price for the trip if the multiplier was 1, and b is a \$2.30 *booking fee*.

Steps 2 and 4 provide exogenous variation from which I identify the response to prices. Since the surge multiplier is a deterministic function of the vector of recommended multipliers $\tilde{\mathbf{m}}_t$, any correlation between demand shocks and prices must come through $\tilde{\mathbf{m}}_t$.²⁵ Once I control for $\tilde{\mathbf{m}}_t$, the residual variation, which arises solely from rounding, is uncorrelated with demand shocks. Supplemental Appendix 4.2 shows in the data that the residual variation is indeed uncorrelated with observables.

Using a regression discontinuity design, Cohen et al. (2016), too, exploit rounding to estimate Uber demand. I rely on stronger assumptions (which I state formally in Section 5) that allow me to capture variation beyond the immediate neighborhood of the discontinuities; otherwise I would not have enough power.

Rounding generates small variation in prices. This might seem problematic given that my goal is to estimate agents' response to larger changes in prices that are generated by changes in pricing policies. However, I observe variation at different price levels: sometimes the multiplier is rounded to 1 or 1.2, but it is also often rounded to 2.2 or 2.3. My estimation procedure averages out these small responses along the whole demand curve to find the response to large price changes without extrapolating. It requires the assumption that, across different times and locations, the shape of the demand curve stays the same (conditional on covariates x_i), but it allows for parallel shifts of the whole demand curve.

²⁵To be precise, I define $\tilde{\mathbf{m}}_t$ to be the vector of all recommended multipliers at times t and $t - 1$. It includes recommended multipliers at time $t - 1$ because surge multipliers are also a function of the recommended multipliers from the previous period due to temporal smoothing.

Response to Pickup Times

The pickup time that riders observe depends on the location of nearby drivers. The number of nearby drivers is correlated with demand shocks: if more people request trips, there are fewer available drivers, which increases pickup times. But drivers’ exact location relative to riders is unlikely to be correlated with demand shocks. For instance, two riders may be half a block apart. If a driver can pick up one of them right away but has to go around the block to pick up the second one, their pickup times could be one and three minutes, respectively.

I use this variation to estimate the response of riders to pickup times. To do this, I include location-by-time-period fixed effects. In essence, my identification strategy compares pairs of riders who are in the same location l and in the same time period t . Any systematic demand shocks that might cause endogeneity would affect both riders equally, so they would be captured by the fixed effects.²⁶ This strategy implicitly assumes that the fixed effects are additive.

Residuals and Value of Time

Figure 4 shows the raw variation in the data from which I identify riders’ response to prices and pickup times. Figure 4a shows that riders are less likely to request trips when the surge multiplier was rounded up than when it was rounded down. To determine whether the multiplier was rounded up or down, I control for the *unrounded multiplier*, an estimate of what the multiplier would have been if there was no rounding in the surge pricing algorithm. I explain how I compute this in Supplemental Appendix 3.1. Figure 4b shows that, within location by time period, riders are less likely to request trips when the pickup time is higher.

Comparing the slopes in Figure 4 gives a sense of how riders trade off prices and pickup times. I compute the ratio between both slopes to obtain the average value of time for

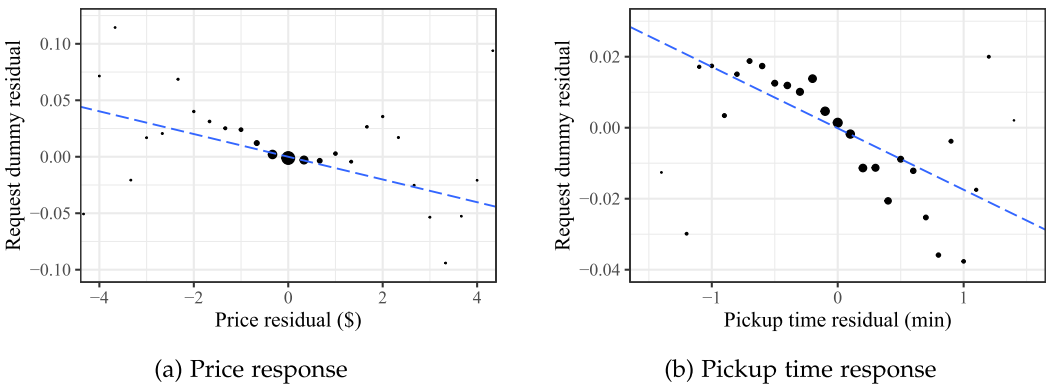


FIGURE 4.—Residuals used to identify the response of riders to prices and pickup times. *Note:* Both figures are binscatters in which the vertical axis is the residual of a dummy for trip request. The point size represents the number of observations. Dashed lines represent a linear fit. Subfigure (a) shows residuals from regressions on the unrounded multiplier; the variable in the horizontal axis is the trip price. Subfigure (b) presents residuals from regressions on location-by-period fixed effects; the variable in the horizontal axis is the pickup time.

²⁶One potential confounder is the distance to major streets. However, if I control for how accessible the rider’s location is from streets where drivers tend to be, I obtain almost identical results.

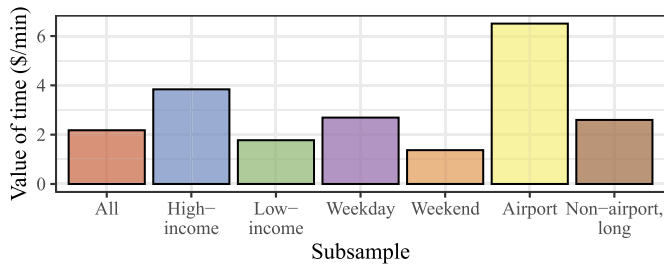


FIGURE 5.—Average value of time for subsamples of the data. *Note:* Each bar represents the average value of time for a subsample of the data. For each subsample, I estimate a linear regression of trip request on price and pickup time. I control for the unrounded price and for the average pickup time by location by time period. I allow coefficients to vary linearly in the base fare (a proxy for trip distance). I also include location-by-hour-of-the-week fixed effects. For each observation, the value of time is the pickup time coefficient divided by the price coefficient.

different subsamples of the data (Figure 5). It is around \$2 per minute for the whole sample. This number is high, but it is comparable with estimates in the literature.²⁷ There is significant variation across subsamples, following the patterns one would expect: it is higher for high-income riders, during the week than during weekends, and it is especially high for airport trips.

Several reasons might lead to the high estimated value of time. First, riders tend to request trips during time-sensitive moments. They might want to be on time for an appointment, or, in the extreme case, they might want to get to the airport in time for a flight. Second, because some trips are requested by more than one rider, the relevant value is the sum of the value of time for all riders. Third, business passengers who do not pay for the trip might have a limited response to prices.²⁸

4.2. Short-Run Supply Response

In the driver movement model, the main parameter I want to estimate is δ_d , how responsive movements are to mean future earnings. This is challenging because movement shocks affect multipliers, which induce changes in mean future earnings.

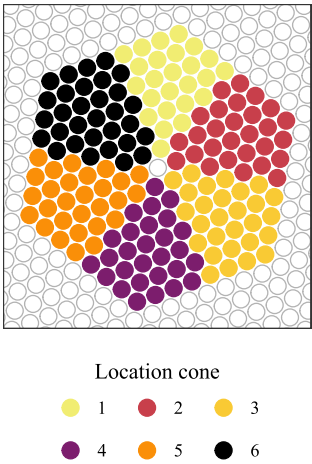
As when estimating demand, I exploit exogenous variation from rounding in the surge pricing algorithm. I now show broad patterns in the data. I aggregate the space surrounding every available driver into six direction cones, as in the figure to the left of Table I. I then run six regressions. In each, I regress a dummy for whether the driver moved to one of these cones on the average multiplier in every one of the six cones. I control for the average recommended multiplier in each cone, and so my estimates are identified from variation that arises from rounding.

Table I shows the estimates from these regressions. The coefficients on the diagonal are all positive and significant. Off-diagonal terms are noisier, but they tend to be negative.

²⁷The value of time is 4.6 times the mean wage in Houston of \$26 per hour, and is similar to the values implied by Cohen et al. (2016) in other cities in the U.S. For Prague, Buchholz et al. (2022) find values that are on average 1.4 times the average wage. In a field experiment, Kreindler (2023) estimates a value of time for commuters in India that is 5 times the average wage.

²⁸These values could also be driven by behavioral effects if riders overreact to pickup times. This, as well as unresponsive business travelers, complicate the interpretation of welfare estimates. Appendix E shows that the main results hold if the true value of time is lower than measured.

TABLE I
EFFECT OF MULTIPLIERS ON MOVEMENT DIRECTION.



	<i>Dependent variable:</i>					
	Dummy for moving to cone in parentheses					
	(1)	(2)	(3)	(4)	(5)	(6)
Avg. multiplier in cone 1	0.22 (0.06)	−0.04 (0.06)	−0.16 (0.05)	0.07 (0.06)	−0.04 (0.05)	−0.04 (0.05)
Avg. multiplier in cone 2	−0.04 (0.06)	0.29 (0.07)	0.08 (0.05)	−0.18 (0.06)	−0.07 (0.06)	−0.07 (0.06)
Avg. multiplier in cone 3	−0.05 (0.06)	−0.09 (0.07)	0.16 (0.06)	−0.04 (0.07)	−0.01 (0.06)	0.04 (0.06)
Avg. multiplier in cone 4	−0.09 (0.05)	−0.05 (0.06)	−0.05 (0.06)	0.37 (0.08)	−0.06 (0.07)	−0.12 (0.06)
Avg. multiplier in cone 5	−0.08 (0.05)	0.04 (0.06)	−0.05 (0.05)	−0.15 (0.07)	0.22 (0.07)	0.01 (0.06)
Avg. multiplier in cone 6	0.02 (0.05)	−0.13 (0.06)	0.03 (0.04)	−0.07 (0.06)	−0.04 (0.06)	0.19 (0.06)
Observations	645,133	645,133	645,133	645,133	645,133	645,133

Note: The figure shows how I aggregate the space surrounding a driver into six cones. In the table, each column represents a regression of a dummy for whether a driver moved to one cone on the average multiplier in every cone. I control for the average recommended multiplier by cone, for the multiplier at the driver’s location, and for location-by-hour-of-the-week fixed effects. Standard errors are clustered by location and by hour of the week.

As multipliers increase in one cone, drivers are more likely to move towards that cone and less likely to move to the other five cones. This is evidence that drivers tend to move towards areas that have high multipliers.

Drivers also respond to high surge multipliers by staying longer. Using an identification strategy similar to the one employed in Table I, I show in Appendix D that drivers are less likely to stop working when the surge multiplier is high.

4.3. Long-Run Response

I calibrate the parameters that determine the long-run elasticities of demand and supply (σ and ρ) to match experimental elasticity estimates.

I estimate the long-run elasticity of demand based on experiments that Uber ran in 2017 in five Latin American cities where Uber was the dominant platform. A random sample of riders was given a 10% or 20% discount for all trips during one week. I find a demand elasticity of 0.633, with a robust standard error of 0.059. Supplemental Appendix 4.4 provides further details about the experiment and estimation.

I base my supply elasticities on the experimental estimates [Caldwell and Oehlsen \(2022\)](#) measure for Houston in spring 2017. They offered 25%–39% higher earnings for one week to a random sample of Uber drivers. They then measured supply elasticities, with separate results by gender and for full-/part-time drivers. One challenge is that the elasticities they measure do not map one-to-one to the long-run elasticity I need. Supplemental Appendix 4.5 discusses how I obtain from their estimates numbers that are appropriate for my model. I use elasticities of 0.3 for full-time men, 0.5 for part-time men, 0.6 for full-time women, and 1 for part-time women.

Using these experimental estimates for my model presents several challenges. First, the demand elasticity was not measured in Houston. Second, these experiments measure the response to one-week changes in prices and earnings, but it would be better to use longer-run elasticities. Third, they measure the response of people who had an Uber account, but not the effect of new riders and drivers. Because of these concerns, Section 6 presents robustness checks in which I run my main counterfactuals using values of σ and ρ that correspond to a range of long-run elasticities. The above concerns suggest that I might want to use higher elasticities. Furthermore, estimates of similar long-run elasticities in the literature are similar to my baseline numbers or higher.²⁹ I thus use demand elasticities ranging from 0.633 to 1.5, and I scale up the baseline supply elasticities by factors between 1 and 2.

5. ESTIMATION AND RESULTS

5.1. Demand

5.1.1. Trip Request

I estimate the trip request model following the ideas from Section 4.1: I identify the price response from rounding in surge multipliers and the pickup time response from within location by time period variation. I estimate the following equation:

$$U_i = \alpha(x_i, l, h) + \beta(x_i)p_i + \gamma(x_i)w_i + g(\tilde{\mathbf{m}}_i, x_i, w_{it}^0; l, h) + \eta_i, \quad (11)$$

where $w_{it}^0 = \mathbb{E}[w_i | l, t]$ is the expected pickup time by location and period. This is the original utility specification (1), but I decompose the error as $\epsilon_i = g(\tilde{\mathbf{m}}_i, x_i, w_{it}^0; l, h) + \eta_i$, where $g(\cdot)$ is a flexible function that controls for recommended multipliers, rider covariates, expected pickup time, location, and hour of the week.

²⁹Other demand estimates include those by [Rosaia \(2023\)](#) for New York (around 1) and an elasticity of 1.2 that Uber typically assumes for the U.S., based on experiments. On the supply side, other estimates include those by [Angrist, Caldwell, and Hall \(2021\)](#), who infer an elasticity of 1.2 for Boston, and [Farber \(2015\)](#), who estimates an elasticity around 0.6 for New York taxi drivers. In another setting with flexible work hours, [Fehr and Goette \(2007\)](#) measure values around 1.2.

I assume that the error η_i is orthogonal to $(p_i, w_i, \tilde{\mathbf{m}}_t, x_i, w_{it}^0, l, h)$. That is true if $g(\cdot)$ captures all the correlation between ϵ_i and covariates, which is justified by the following intuition. First, p_i is fully determined by (i) recommended multipliers, (ii) the base fare, and (iii) rounding in the surge algorithm. Thus, if one controls flexibly for recommended multipliers and the base fare, the only variation in prices that remains comes from exogenous rounding. Second, if one controls for w_{it}^0 , only the variation in w_i within location by period remains, which, as argued in Section 4.1, is exogenous because it is driven by riders' exact location relative to drivers.

I now formalize this argument. I rely on the following assumption:

ASSUMPTION 1: If rider i takes a trip, her utility is $U_i = \alpha(x_i, l, h) + \beta(x_i)p_i + \gamma(x_i)w_i + \epsilon_i$, which satisfies:

1. $w_i = w_{it}^0 + \xi_i$, where $w_{it}^0 = \mathbb{E}[w_i | l, t]$, and ξ_i is orthogonal to $(\epsilon_i, x_i, \tilde{\mathbf{m}}_t)$.
2. For every (l, h) , the support of $(\tilde{\mathbf{m}}_t, w_i, w_{it}^0, x_i)$ is a full-dimensional, compact set $\mathcal{S}$.
3. $p_i = p(\tilde{\mathbf{m}}_t; \bar{p}_i, l)$, where $p(\cdot)$ is a deterministic function that, for all $(\bar{p}_i, l)$, has at least one nonremovable discontinuity in the interior of $\mathcal{S}$.
4. $\mathbb{E}[\epsilon_i | \tilde{\mathbf{m}}_t, x_i, w_{it}^0, l, h]$ is Lipschitz continuous in $\tilde{\mathbf{m}}_t$.³⁰
5. ϵ_i has full support in $\mathbb{R}$.

Part 1 states that variation in pickup times within location by period ($w_i - w_{it}^0$) is orthogonal to demand shocks, rider covariates, and recommended multipliers. I argued in Section 4.1 that this assumption holds. Part 3 simply states a property of the surge pricing algorithm: the fare p_i is a deterministic function of recommended multipliers and the base fare that has discontinuities in $\tilde{\mathbf{m}}_t$. Part 4, which states that demand shocks and recommended multipliers are related smoothly, is satisfied because the surge pricing algorithm computes recommended multipliers as smooth functions of market observables. This assumption, which is necessary to be able to distinguish discontinuities in recommended multipliers from rounding, is analogous to the assumption in a regression discontinuity design that conditional means of the outcome variable are continuous in the forcing variable. Finally, parts 2 and 5 are technical assumptions about the support of variables.

The following proposition is the main identification result:

PROPOSITION 1: *Under Assumption 1, rider i 's utility can be rewritten as equation (11), where $g(\tilde{\mathbf{m}}_t, x_i, w_{it}^0, l, h) = \mathbb{E}[\epsilon_i | \tilde{\mathbf{m}}_t, x_i, w_{it}^0; l, h]$ and $\mathbb{E}[\eta_i | p_i, w_i, \tilde{\mathbf{m}}_t, x_i, w_{it}^0, l, h] = 0$.*

The coefficients $\beta(x_i)$ and $\gamma(x_i)$ are identified.

Based on Proposition 1, I can estimate equation (11) using standard regression techniques: there is no endogeneity after controlling for $g(\tilde{\mathbf{m}}_t, x_i, w_{it}^0; l, h)$. The specification I choose for $g(\cdot)$ is important: it must be flexible enough to capture the true functional form of $\mathbb{E}[\epsilon_i | \tilde{\mathbf{m}}_t, x_i, w_{it}^0; l, h]$, but not so flexible that it captures the discontinuities in the surge multipliers. Given part 4 of Assumption 1, the key is to choose a Lipschitz continuous specification—that is, a specification with bounded derivatives. As I explain below, I use a combination of high-order splines.³¹

³⁰This assumption can be weakened to Hölder continuity but not to uniform continuity.

³¹A specification based on splines is consistent under the assumption that the true function is in the span of the spline. Informally, that is approximately the case if g does not vary too much. One could use sieve methods to avoid this assumption; in that case, a constraint on sieve coefficients α_k of the form $\sum_k b_k \alpha_k < \infty$ is required to ensure the sieve function is Lipschitz continuous.

This model assumes no unobserved heterogeneity in $\beta(x_i)$ and $\gamma(x_i)$.³² My estimates, as well as the welfare effects I measure, should be interpreted as averages conditional on x_i . This requires the assumption that unobserved heterogeneity is not correlated with prices or pickup times.

Functional Form Assumptions. The vector x_i includes a constant, log income, a dummy for occasional riders, the unsurged fare (a measure of traffic-adjusted trip distance), and dummies for trips to the airport, during the weekend, and starting from home. The pickup time coefficient is linear, $\gamma(x_i) = \theta^w x_i$. The price coefficient takes the form $\beta(x_i) = s(\theta^p x_i)$, where $s(\cdot)$ is an increasing function that varies across specifications.

I set $g(\cdot)$ as a sum of high-dimensional splines of $\tilde{\mathbf{m}}_i$, w_{it}^0 , and the *unrounded price* $\hat{p}_i$, the fare the rider would have seen if there was no rounding (see Supplemental Appendix 3.1). I assume the intercept term $\alpha(x_i; l, h)$ is additively separable into a linear function of x_i and a flexible function of (l, h) with 155 degrees of freedom. I plug in $\bar{w}_{it}$, the average pickup time by (l, t) , for w_{it}^0 .³³ See Supplemental Appendix 3.2 for further details.

I assume that the error η_i is distributed i.i.d. logistic. This is, thus, a logit model with nonlinear coefficients. I estimate it by maximum likelihood.

Results. Table II presents estimates of the main parameters θ^p and θ^w . In specification (1), I set $s(\cdot)$ to be the identity function, so that the price coefficient is linear ($\gamma(x_i) = \theta^p x_i$), and I do not include the control function $g(\cdot)$. This means that I use all the variation in prices and pickup times, so the parameters are likely to be biased due to endogeneity. The average price coefficient is negative and highly significant, and the average pickup time coefficient is a precisely measured zero. Specification (2) adds the control function $g(\cdot)$, so that I only use exogenous variation in prices and pickup times. Both average coefficients go down, as one would expect if simultaneity biases the coefficients in (1) towards zero. Standard errors are also larger—especially those of price parameters—because the control function absorbs a large fraction of the variation in prices and pickup times.

One challenge with specification (2) is that the linear functional form $\gamma(x_i) = \theta^p x_i$ results in a few positive price coefficients (for 1.46% of observations). That is problematic for the main model because the value of time is then negative and, even worse, it diverges when the price coefficient is close to zero. Specification (3) thus sets $s(x) = -\exp(-x)$ to constrain the price coefficient to be negative. The two negative signs ensure that $s(\cdot)$ is increasing. The overall distribution of the price coefficient is almost the same as in (2): its average and standard deviation are -0.050 and 0.0193 , and they are -0.0489 and 0.0217 in (2). The heterogeneity parameters all have the same sign, and the pickup time parameters are virtually unchanged.

The specification I use in the main model is (3). I now present some patterns implied by that specification. The average value of time across all sessions is \$2.25 per minute, which is somewhat higher than the values from Section 4.1. The heterogeneity parameters imply the patterns one would expect. As Figure 6a shows, the value of time is higher for high-income riders, during weekdays, and when people are going far (i.e., when the base fare is

³²Buchholz et al. (2022) find substantial unobserved heterogeneity. Drivers in my data make a binary choice, so I do not observe substitution elasticities that I could use to identify heterogeneity.

³³There are few observations by (l, t) , so one might worry that $\bar{w}_{it} \rightarrow w_{it}^0$. An alternative specification (a Bayesian hierarchical model) that avoids this problem results in very similar estimates.

TABLE II
ESTIMATES OF PARAMETERS OF THE TRIP REQUEST MODEL.

	Specification:					
	No control function, linear price coefficient		With control function, linear price coefficient		With control function, exponential price coefficient	
	(1)		(2)		(3)	
	Price coefficient	Pickup time coefficient	Price coefficient	Pickup time coefficient	Price coefficient	Pickup time coefficient
Constant	−0.0263 (0.0026)	−0.0008 (0.0094)	−0.0489 (0.0155)	−0.0964 (0.0138)	3.08 (0.2531)	−0.0964 (0.0136)
Occasional	0.0223 (0.0012)	0.0188 (0.0076)	−0.0147 (0.0217)	0.0501 (0.0168)	−0.121 (0.0778)	0.0501 (0.0170)
Log income	−0.0096 (0.0010)	0.0022 (0.0038)	0.0399 (0.0261)	−0.0002 (0.0244)	0.414 (0.0551)	−0.0002 (0.0232)
Base fare	−0.0001 (0.0001)	−0.0006 (0.0004)	0.0016 (0.0009)	−0.0002 (0.0013)	0.0296 (0.0286)	−0.0002 (0.0012)
Airport	0.0041 (0.0034)	0.0367 (0.0108)	0.0185 (0.0242)	0.0295 (0.0384)	0.573 (0.0679)	0.0294 (0.0415)
Weekend	0.0019 (0.0018)	0.0145 (0.0102)	0.0009 (0.0205)	0.0394 (0.0251)	0.258 (0.1024)	0.0393 (0.0250)
Home	0.0062 (0.0027)	−0.0074 (0.0129)	0.0120 (0.0427)	0.0431 (0.0438)	0.904 (0.0342)	0.0432 (0.0443)

Note: This table presents estimates of the trip request model. Every specification is presented in two columns; the first column shows estimates of θ^P (the parameters of the price coefficient $\beta(x_i)$), while the second column shows estimates of θ^w (the parameters of the pickup time coefficient $\gamma(x_i)$). Covariates x_i are demeaned, so the constant terms correspond to the values $\theta^P x_i$ and $\theta^w x_i$ taken at the average value of x_i . Specification (1) has a linear price coefficient $\gamma(x_i) = \theta^P x_i$, and it does not include the control function $g(\cdot)$. Specification (2) also has a linear price coefficient, but it does include the control function. In specification (3), the price coefficient takes the form $\gamma(x_i) = -\exp(-\theta^P x_i)$, and I also include the control function. Standard errors are computed using a sandwich estimator with two-way clustering by location and hour of the week.

high), especially to the airport.³⁴ Note, however, that some heterogeneity parameters are somewhat imprecisely estimated.

Figure 6b shows the distribution of the price elasticity for different types of riders. Occasional riders are more elastic. Among frequent riders, high-income riders have lower elasticities. Riders who take trips from home are about half as elastic as riders who take trips from other places. The average elasticity for the whole sample is 0.268. Other ride-hailing papers find short-run demand elasticities between 0.35 and 1.³⁵ Those studies focus on cities with better alternative transportation options than Houston, so one should expect them to find higher elasticities.

5.1.2. Opening the App

In the rider arrival model (Section 3.1.2), I need to find values for ρ , which measures the long-run elasticity of demand, and for the demand shifters A_{lhx} . I set their values jointly

³⁴The value of time for trips from home is around 50% higher than in Figure 6a. For occasional riders, it is \$0.60–\$1.52 per minute, which resembles that of the lowest-income frequent riders.

³⁵Cohen et al. (2016) find elasticities between 0.35 and 1 using pooled data from New York, San Francisco, Chicago, and Los Angeles. Rosaia (2023) measures elasticities around 1 in New York.

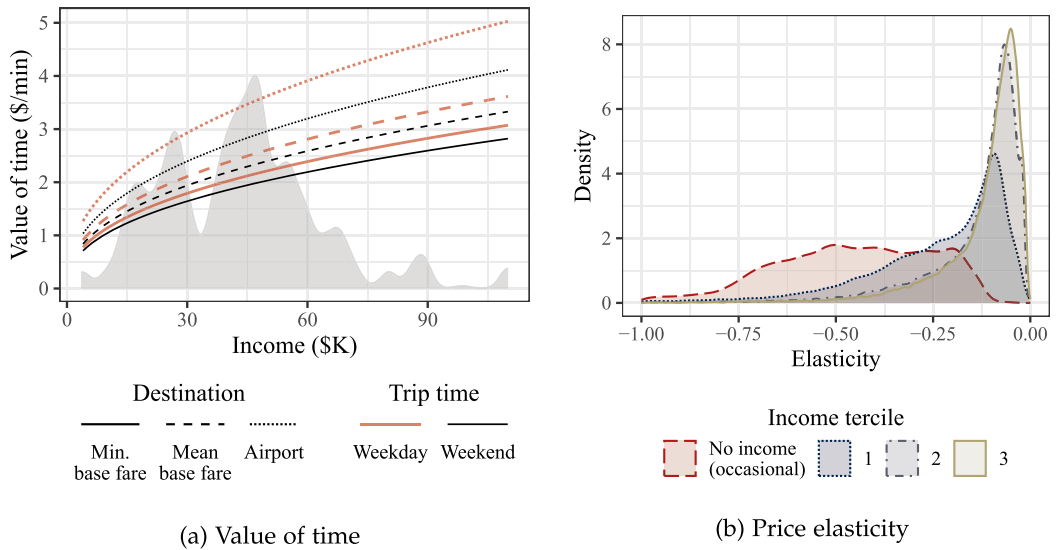


FIGURE 6.—Value of time and price elasticity from the trip request model. *Note:* Subfigure (a) shows how the value of time varies with income, trip distance, and time of the week for trips that do not start from the rider's home. The thick, gray lines (red online) represent weekday trips; the thin, black lines represent weekend trips. The line type (solid, dotted, or dashed) represents the type of destination. The minimum base fare is \$5.41, and the mean base fare is \$10.65. The income distribution is shown in gray. Subfigure (b) shows the density of the price elasticity for occasional riders as well as for frequent riders by income tercile.

so that (a) for every (l, h, x) , the observed arrival rate is equal to the predicted arrival rate given the utilities U_{lhx} from the data, and (b) the market-wide elasticity of demand is equal to the value of -0.633 that I estimate from demand experiments (Section 4.3).³⁶ This results in a value of $\rho = 1.79$.

A few trips that were requested from external apps (such as Google Maps) are not included in the main demand data set. To account for them, I allow for a multiplicative scale factor ψ^d that shifts all of demand up or down. In Section 5.2.3, I describe a similar scale factor ψ^s for supply. I set both factors jointly such that simulations of the market equilibrium in the status quo result in the same number of trips and the average surge multiplier as in the data. This results in $\psi^d = 1.013$.

5.2. Supply

I estimate the supply model allowing for heterogeneity across four driver types d that are defined by gender and whether drivers are classified as full- or part-time.

³⁶To estimate U_{lhx} consistently, I aggregate combinations of (l, h, x) into larger groups because there are too many values of l and h and because x is continuous (see Supplemental Appendix 3.3).

5.2.1. Movement

In the driver movement model from Section 3.2.1, I identify δ_d from variation that arises from rounding in surge multipliers. The model I estimate is

$$\Pr(l_{j,t+1} = k | \mathbf{s}_t) = \frac{\exp(\omega_{lkhd} + \delta_d v_{dk}(\mathbf{s}_t) + g^M(\tilde{\mathbf{m}}_{tk}; l, h, d))}{\sum_{k'} \exp(\omega_{lkhd} + \delta_d v_{dk'}(\mathbf{s}_t) + g^M(\tilde{\mathbf{m}}_{tk'}; l, h, d))}. \quad (12)$$

This is equation (5), except that the unobserved term ζ_{kt} is now equal to $g^M(\tilde{\mathbf{m}}_{tk}; l, h, d)$, a flexible, smooth function of the recommended multipliers surrounding location k . This allows me to estimate δ_d from rounding. The intuition is similar to that in the trip request model. Mean future earnings $v_{dk}(\mathbf{s}_t)$ are a function of multipliers, which, in turn, are a function of recommended multipliers. If g^M is flexible enough, it captures all of the variation in $v_{dk}(\mathbf{s}_t)$ that arises from recommended multipliers, and the residual variation that identifies δ_d comes solely from rounding.

I now formalize this argument, which relies on the following assumption:

ASSUMPTION 2: Driver movements are determined by $l_{j,t+1} = \arg\max_k y_{j,t+1}^k$, where $y_{j,t+1}^k = \omega_{lkhd} + \delta_d v_{dk} + \zeta_{j,t+1}^k$. This equation satisfies the following conditions, where $\mathbf{m}_{tk}$ and $\tilde{\mathbf{m}}_{tk}$ are vectors of multipliers relative to location k :

1. For every h , the support of $\tilde{\mathbf{m}}_{tk}$ is a full-dimensional, compact set $\mathcal{S}^{\tilde{m}}$.
2. $v_{dk} = \tilde{v}_{dkh}(\tilde{\mathbf{m}}_{tk}) = v_{dk}(\mathbf{m}_{tk}(\tilde{\mathbf{m}}_{tk}); h)$, where $\tilde{v}_{dkh}$ is a function that has at least one nonremovable discontinuity in the interior of $\mathcal{S}^{\tilde{m}}$ for all (d, k, h) .
3. $\mathbb{E}[\zeta_{j,t+1}^k | \tilde{\mathbf{m}}_{tk}; l, h, d]$ is Lipschitz continuous in $\tilde{\mathbf{m}}_{tk}$.
4. $\psi_{jt}^k = \zeta_{jt}^k - \mathbb{E}[\zeta_{j,t+1}^k | \tilde{\mathbf{m}}_{tk}; l, h, d]$ are i.i.d. and follow a standard Gumbel distribution.

This assumption reframes the movement rule as a latent variable model. Part 1 is a technical assumption about full support of the recommended multipliers. Part 2 provides the variation I use to identify δ_d . It holds because the discontinuities in the surge pricing algorithm induce discrete changes in mean future earnings. Part 3 is entirely analogous to part 4 of Assumption 1 for demand estimation. Lastly, part 4 is a standard functional form assumption that results in closed-form expressions for probabilities, such as with multinomial logit models.

I now state the main identification result for the movement model:

PROPOSITION 2: Under Assumption 2, movement probabilities are given by equation (12), where $g^M(\tilde{\mathbf{m}}_{tk}; l, h, d) = \mathbb{E}[\zeta_{j,t+1}^k | \tilde{\mathbf{m}}_{tk}; l, h, d]$. The coefficients δ_d are identified.

Based on Proposition 2, I can estimate equation (12) by maximum likelihood: there are no endogeneity issues after including $g^M(\cdot)$. Choosing the specification for g^M involves the same issues discussed in Section 5.1.1.

Implementation Details. Although the model is fully flexible across driver types, I do not have enough power to estimate δ_d and $v_{dk}(\cdot)$ type by type. Thus, I estimate one single δ and $v_k(\cdot)$ for all types. This means that the heterogeneity in welfare effects that I measure in Section 6.3 does not account for additional differences that might arise if some types of drivers are more likely to chase high surge multipliers.

TABLE III
ESTIMATES OF SUPPLY PARAMETERS.

Panel A: Driver movement model								
Parameter	Model without control function (1)				Model with control function (2)			
δ	0.190 (0.014)				0.0904 (0.016)			
Panel B: Driver exit model								
Parameter	Model without control function				Model with control function			
	Driver type				Driver type			
	Full-time men (1)	Part-time men (2)	Full-time women (3)	Part-time women (4)	Full-time men (5)	Part-time men (6)	Full-time women (7)	Part-time women (8)
κ_d	1.63 (0.13)	1.08 (0.09)	1.82 (0.11)	0.91 (0.08)	0.97 (0.05)	1.06 (0.05)	1.99 (0.47)	1.03 (0.08)
Panel C: Driver entry model								
Parameter	Driver type							
	Full-time men (1)		Part-time men (2)		Full-time women (3)		Part-time women (4)	
σ_d	0.152		0.186		0.234		0.357	

Note: This table presents estimates of the main supply parameters. Panel A presents estimates of δ from the movement model, both when I include the control function and when I do not include it. Standard errors are bootstrapped. Panel B presents estimates of κ_d from the driver exit model. I consider a specification that does not include the control function (columns (1)–(4)) and one that includes it (columns (5)–(8)). Standard errors are computed using a sandwich estimator with two-way clustering by driver and hour of the week. Panel C presents the calibrated values of σ_d in the entry model. I do not compute standard errors because these values come from a calibration.

I set $v_k(\cdot)$ to be the beliefs that arise from the data. To compute them, I run model (9)—which I use to compute beliefs from simulated histories—on the actual data. Then I plug $v_k(s_t)$ into every observation.

I set $g^M(\tilde{\mathbf{m}}_{tk}; l, h, d)$ to be a sum of high-dimensional splines of recommended and unrounded multipliers in the six nearest hexagon rings. I set ω_{lkhd} to be the sum of three terms that capture, respectively, road patterns, traffic patterns, and the tendency of different types of drivers to work in certain areas at certain times. I estimate the model by maximum likelihood. See Supplemental Appendix 3.5 for details.

Results. Panel A of Table III presents estimates of δ . Column (1) shows that I find a positive and significant coefficient if I do not include $g^M(\cdot)$. However, this specification identifies δ from all the variation in v_k ; thus, endogeneity is likely to be a problem. Simultaneity can bias this coefficient towards zero. Some omitted variables may, instead, bias δ away from zero, such as unobservable information about demand shocks that induce drivers to move towards high-multiplier regions.³⁷

³⁷This would occur if drivers know of the end of a concert and move towards the area. This biases δ upwards if drivers respond to information beyond what can be inferred from surge multipliers.

Column (2) presents my main estimate. I include $g^M(\cdot)$ to identify δ only from variation from rounding. The estimate I obtain is positive and significant, but it is smaller than in column (1)—which suggests that omitted variable bias was a larger problem than endogeneity. To interpret this estimate, consider a driver who is equally likely to move to one of four destinations surrounding him in the next period. If earnings in one of the four locations increase by \$3, which roughly corresponds to an increase in surge multipliers from 1 to 1.5 in all surrounding hexes, the probability that the driver goes in that direction increases from 0.25 to 0.304.³⁸

5.2.2. Exit

To estimate the exit model (Section 3.2.2), I exploit rounding in surge pricing to identify κ_d , which determines how likely drivers are to extend their shift.

Driver j decides to stay at time t if $v_{d,0}^s(\tilde{s}_{jt}) - s_{jt} > 0$. I assume that the error s_{jt} is distributed i.i.d. logistic with scale a_d , so the probability of staying is $\Lambda(\frac{1}{a_d}v_{d,0}^s(\tilde{s}_{jt}))$, where $\Lambda(\cdot)$ is the standard logistic function. I assume that the driver-specific duration T_j is distributed gamma with parameters $(\alpha_{dh}, \beta_{dh})$ that vary by driver type and hour of the week. The likelihood of observing driver j 's decisions whether to stay or exit are thus given by

$$\mathcal{L}_j = \int \underbrace{\left[1 - \Lambda\left(\frac{1}{a_d}v_{d,0}^s(\tilde{s}_{jt_j})\right)\right]}_{\text{Prob. of leaving at time } \tilde{t}_j} \underbrace{\prod_{t \in \mathcal{T}_j} \Lambda\left(\frac{1}{a_d}v_{d,0}^s(\tilde{s}_{jt})\right)}_{\text{Prob. of staying at time } t} dF^T(T; \alpha_{dh}, \beta_{dh}), \quad (13)$$

where $\mathcal{T}_j$ is the set of periods when the driver could have exited but stayed, $\tilde{t}_j$ is the period when he eventually exited, and F^T is a gamma distribution. I estimate this model by maximum likelihood, integrating by Gauss–Laguerre quadrature.

I let $v_{d,0}^s(\cdot)$ be the beliefs that are consistent with the data. I compute them by following the same process I use in Section 3.5.1 to estimate beliefs from simulated histories, but I run it on the actual data.

The model above identifies κ_d from variation in multipliers, which could be endogenous. I adjust the model to use only variation that arises from rounding (see Supplemental Appendix 3.6). Similarly to how I estimate the movement model, I assume that the error can be written as $s_{jt} = g^E(\tilde{\mathbf{m}}_{tk}; l, h, d) + \vartheta_{jt}$, where ϑ_{jt} is an i.i.d. error.

Results. Panel B of Table III presents estimates for κ_d . Results change little whether I include the control function $g^E(\cdot)$ or not, which suggests that endogeneity is not a major concern. Focusing on my main specification (columns (5)–(8)), the estimates of κ_d for men and part-time women imply that if they were guaranteed constant earnings that are higher by \$5 per hour, they would extend their shift by around five minutes. Full-time women are less flexible: they would extend their shift by 2.5 minutes. The estimates for α_{dh} and β_{dh} predict shift lengths with a distribution that resembles the empirical distribution closely (see Supplemental Appendix 4.3).

³⁸Lu, Frazier, and Kislev (2018) estimate a similar multinomial logit model from a surge outage that affected drivers who use iOS but not those who use Android. Based on their estimates, the example above shifts the probability from 0.25 to 0.269. This smaller change is likely driven by the fact that their model has no fixed effects; consequently, the error term captures a lot of predictable variation.

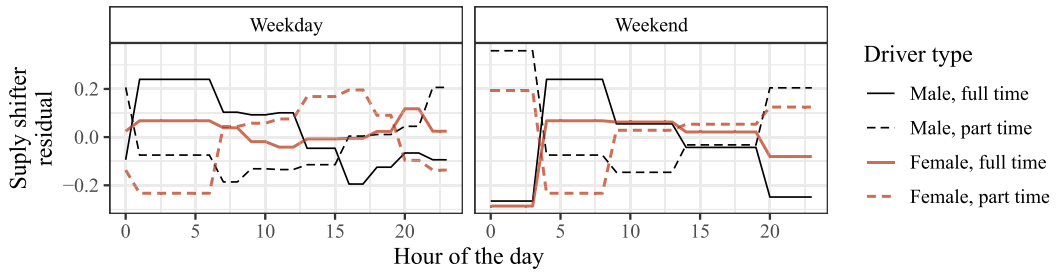


FIGURE 7.—Heterogeneity in the propensity to work at different times. *Note:* This figure shows which driver types are disproportionately likely to work at different times of the week. I average supply shifters B_{lhdT} at the driver type by hour-of-the-week level. I plot the residuals of these means from regressions on driver type plus hour-of-the-week fixed effects. The vertical scale is in logs. These plots show piecewise constant functions because I aggregate hours of the week into 14 groups to compute B_{lhdT} (see Supplemental Appendix 3.3).

5.2.3. Entry

The parameters of the driver entry model are σ_d , which determine long-run supply elasticities, and supply shifters B_{lhdT} . I set both sets of parameters jointly so that (a) the market-wide elasticity of supply for every driver type d is consistent with the values I take from [Caldwell and Oehlsen \(2022\)](#), (b) for every (l, h, d) , the observed arrival rate is equal to the predicted arrival rate given the expected utilities V_{lhdT} that arise from earnings in the data, and (c) for every (h, d) , the distribution of driver-specific durations T_j is consistent with the $\text{Gamma}(\alpha_{dh}, \beta_{dh})$ distribution I estimate in Section 5.2.2. See Supplemental Appendix 3.3 for additional details.

Panel C of Table III presents the values I obtain for the elasticity parameters σ_d . They move in line with the elasticities from [Caldwell and Oehlsen \(2022\)](#). The supply shifters B_{lhdT} , which capture the propensity to work (holding the expected utility fixed), follow the general patterns one would expect. They are higher during the day, which is when many drivers are working. They are also higher for part-time drivers and men, who account for a larger fraction of shifts than full-time drivers and women. Figure 7 shows that there is also substantial variation beyond these general trends. Relative to a typical driver, full-time drivers are less likely to work during bar hours, and they are more likely to work during the early morning—especially men. Part-time men are especially likely to work during bar hours, and part-time women are very prone to working during the afternoon and evening on weekdays. As we will see in Section 6.3, these patterns are the main reason why surge pricing has different welfare effects on different types of drivers.

I also allow for a multiplicative scale factor ψ^s that shifts all of supply up or down, similar to the demand scale factor ψ^d . This corrects for the fact that some drivers work not only for UberX, but also for UberBlack, UberXL, or UberEats, but my data set only has drivers that mostly do UberX. As described in Section 5.1.2, I set ψ^s and ψ^d jointly so that the number of trips and the average surge multiplier are consistent between simulations and the data. This results in $\psi^s = 1.076$.

5.3. Matching

I estimate two elements of the matching model: the distribution of pickup times $G(\cdot | l_i, l_j, b_j, h)$ for rider-driver pairs and the acceptance probabilities ϕ^a and ϕ^b .

For pickup times, I first fit a random forest to obtain a prediction $\hat{w}(X_i, X_j, h)$ of pickup times as a function of rider and driver coordinates (X_i and X_j) and the hour of the week. I

also fit a linear model of the standard deviation of the residual of this model as a function of the prediction $\hat{w}$. Let $\hat{sd}(\hat{w})$ be the prediction from this model. To generate draws from $G(\cdot|l_i, l_j, b_j, h)$, I first draw X_i and X_j from the empirical distributions of all pickup coordinates in location l_i and all coordinates of available drivers in l_j . Then I draw the pickup time $w_{ij}^p \sim G(\cdot|l_i, l_j, b_j, h)$ from a lognormal distribution with mean $\hat{w}(X_i, X_j, h)$ and standard deviation $\hat{sd}(\hat{w}(X_i, X_j, h))$.

I fit driver acceptance rates using the method of simulated moments.³⁹ The two moments I match are the mean realized pickup time (3.52 minutes) and the fraction of trips that are assigned to busy drivers (13.7%). These two moments relate to the parameters in an intuitive way: higher acceptance probabilities result in matches with lower pickup times, and higher ϕ^b relative to ϕ^a leads to a higher fraction of trips assigned to busy drivers. I simulate moments from every period in the data, given the trip requests and available drivers: for parameters (ϕ^a, ϕ^b) , I run the matching model and compute simulated moments.

Results

I estimate acceptance probabilities $\hat{\phi}^a = 0.692$ (s.e. = 0.036) and $\hat{\phi}^b = 0.138$ (s.e. = 0.010).⁴⁰ These numbers are close to their empirical counterparts but somewhat lower (see Supplemental Appendix 4.1). The discrepancy arises because I aggregate requests into two-minute batches, which allows some infeasible matches.⁴¹

5.4. Model Fit

In Supplemental Appendix 5.1, I compare the data with simulations of the status quo in equilibrium. The model captures very well the high-resolution spatial patterns in the data as well as hourly patterns of supply and demand. The simulated distribution of surge multipliers is almost identical to the empirical distribution.

6. THE WELFARE EFFECTS OF SURGE PRICING

6.1. Measuring Welfare

Rider and driver surplus are sums for the whole market (over locations, hours of the week, and agent types) of the rider and driver surplus defined in Sections 3.1.2 and 3.2.3:

$$RS = \sum_{lhx} RS_{lhx}, \quad DS = \sum_{lhdT} DS_{lhdT}. \quad (14)$$

I compute Uber's (current) profit as

$$\Pi = \sum_n ((1 - \tau - \nu) p_n - y_n - I_n). \quad (15)$$

³⁹I do not estimate ϕ^a and ϕ^b directly from the data because my priority is to model the distribution of pickup times correctly to capture the magnitude of matching inefficiencies.

⁴⁰Standard errors are computed using a sandwich estimator, computing matrices by simulation.

⁴¹For instance, a rider who requests a trip right after the period starts cannot be matched to a driver that became available right before the period ends. ϕ^a and ϕ^b should thus be interpreted as the acceptance probability times the probability that a match is feasible due to intra-period timing.

In this sum, n indexes trips, p_n is the trip price, y_n is the payment to the driver,⁴² and I_n is insurance costs.⁴³ Uber gets a fraction $(1 - \tau - \nu)$ of the fare, where τ is a 2% sales tax and ν is a 1% credit card transaction cost.

6.2. Welfare Effects of Surge Pricing

In Figure 8, I compare counterfactuals with different surge pricing and uniform pricing policies. The horizontal axis represents the average surge multiplier for the whole market.⁴⁴ In the upper left subfigure, the vertical axis represents total welfare—the sum of rider surplus, driver surplus, profit, and tax revenue—relative to the status quo. The dotted lines cross at the status quo.

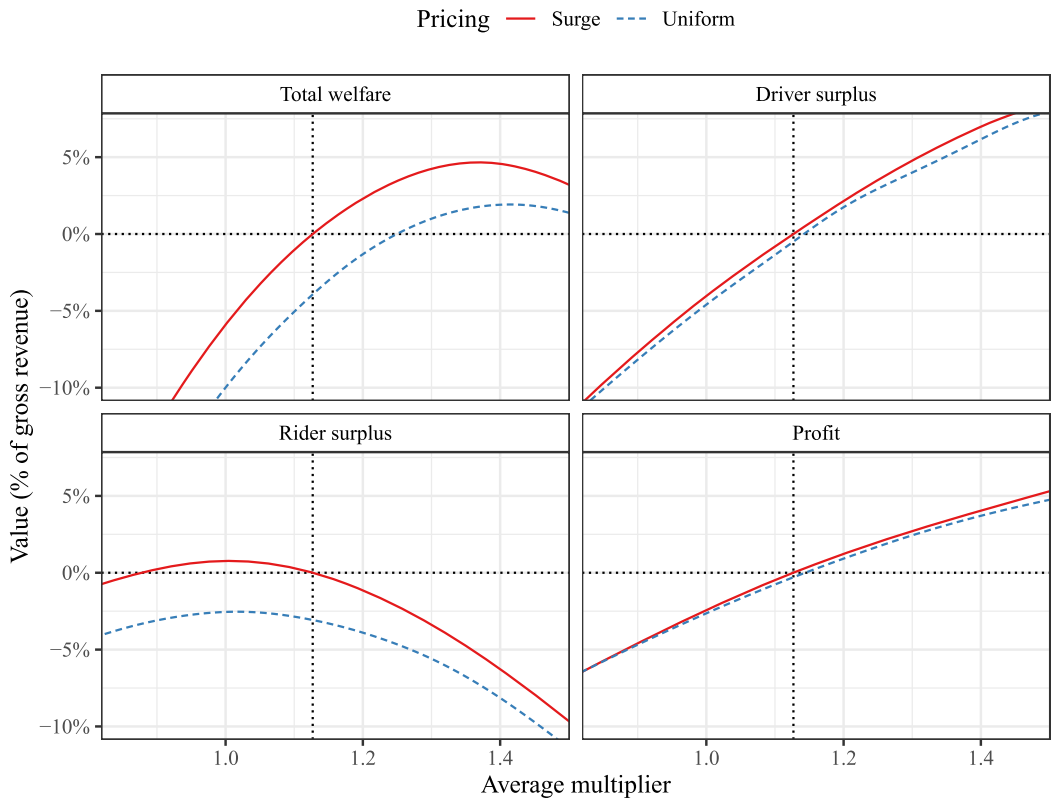


FIGURE 8.—Welfare under different pricing policies. *Note:* These figures compare welfare and its components for different pricing policies. The horizontal axis represents the average surge multiplier. The vertical axis represents total welfare, rider surplus, driver surplus, or profits relative to the status quo. The vertical and horizontal dotted lines highlight the status quo. Curves for surge pricing represent policies in which multipliers are computed as in the status quo, but are scaled up or down by a constant factor. Curves for uniform pricing represent policies that have a unique multiplier for the whole market. All curves were smoothened using a cubic spline with 15 degrees of freedom.

⁴²I use a fixed commission rate of 26.3%, which is the average in the data.

⁴³Uber pays per-mile insurance whenever a driver is picking up or dropping off a rider. I use a cost of \$0.30 per mile, which is somewhat below market rates for private customers in Houston.

⁴⁴Whenever I refer to an average surge multiplier, I refer to an average weighted by the base fare.

The solid line represents alternative surge pricing policies in which the multiplier is scaled up or down by a constant factor: If under certain market conditions the surge multiplier in the status quo is m_{lt} , then under those same conditions the surge multiplier is am_{lt} if the scale factor is a . Increasing a thus entails an increase in average multipliers. The dashed line represents policies in which there is a uniform surge multiplier that applies to all times of the week and all locations. Thus, moving vertically from a uniform pricing policy towards the surge pricing policy right above it represents a mean-preserving spread of surge multipliers.

Surge pricing dominates uniform pricing in the sense that, for every level of the average multiplier, welfare is higher with surge pricing. This is evidence that surge pricing brings efficiency gains. Both for surge and uniform pricing, welfare has an inverted-U shape with a maximum at around 1.4. Uber, therefore, is pricing lower than is socially optimal.

The other three subfigures break down welfare into rider surplus, driver surplus, and profit.⁴⁵ Rider surplus is nonmonotonic because of competing effects on high-income riders, who prefer higher prices because of low pickup times and reliable trips, and low-income and occasional riders, who prefer lower prices. On aggregate, riders would prefer somewhat lower prices than in the status quo, with an average multiplier around 1. Driver surplus and profit are increasing for the whole range of multipliers. This means that Uber prices well below the level for current profit maximization, consistent with Rosaia (2023).

It might be surprising that Uber does not price to maximize profit. However, this model computes profit on a time frame of a few weeks (the time frame of long-run elasticities). Uber's goal is not to maximize its current profit—in fact, it has lost money since it was founded. Instead, it aims to maximize shareholder value. Setting prices at a lower price at which rider surplus is high is the right strategy if Uber believes an important determinant of long-run performance is consumer satisfaction. This is a predominant belief among technology companies.⁴⁶

Welfare Effects With a Fixed Average Multiplier

One remarkable takeaway from Figure 8 is that most of the welfare gap between surge and uniform pricing is accounted for by rider surplus. Driver surplus and profit are also higher for surge pricing, but only slightly. To understand why, Figure 9 breaks down the welfare effect on riders and drivers of moving from uniform to surge pricing if the average multiplier is held constant at the status quo average.^{47,48}

The welfare gains of riders are substantially larger than those of drivers for two reasons. First, surge pricing mitigates supply-demand imbalances, saving market participants' time. Riders wait less time to be picked up, and drivers spend less time waiting to be matched and picking up riders. The bars in Figure 9a measure those time savings. Drivers save around twice as much time per trip as riders.⁴⁹ However, the value of one minute is much higher for riders than for drivers, and so in dollar terms, riders'

⁴⁵I do not show tax revenue, which only accounts for a very small fraction of welfare.

⁴⁶As said by Jeff Bezos, founder and former CEO of Amazon, "take a long-term view, and the interests of customers and shareholders align" (Bezos (2013)).

⁴⁷Figure 9 does not show an additional effect on driver surplus: surge pricing allows drivers to make better decisions about when to stop working. That effect, however, only affects marginal drivers who are close to indifferent between staying or leaving, so it is a second-order effect.

⁴⁸I do not break down the effects on profit, but they are almost exactly proportional to the effects on driver surplus.

⁴⁹On average, surge pricing only saves 7 seconds per trip for riders. Supplemental Appendix 5.4 shows that the main effect is that surge pricing cuts down the upper tail of the pickup time distribution.

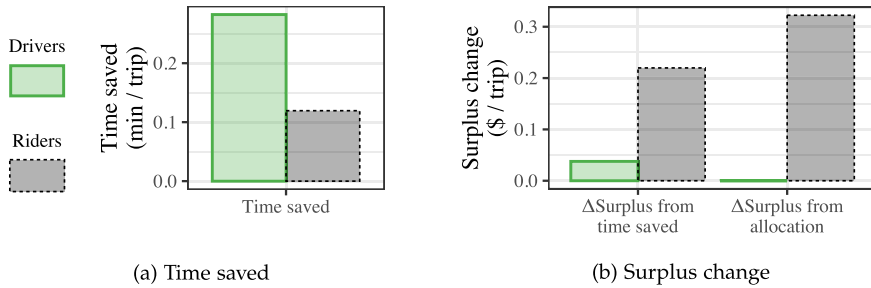


FIGURE 9.—Efficiency gains of surge pricing. *Note:* These figures decompose the welfare effects of moving from uniform to surge pricing while holding the average multiplier constant at the status quo level. Subfigure (a) shows how much time riders and drivers save. Subfigure (b) decomposes the change in rider and driver surplus into time savings (the decrease in pickup time times riders' value of time, and the decrease in idle time times drivers' average hourly wage) and a better allocation (lower wasted surplus due to denied trips).

time savings are more valuable. The bars at the left in Figure 9b measure those welfare gains.

Second, allocative efficiencies (the bars at the right in Figure 9b) only benefit riders. When there are few available drivers, trips are allocated randomly to those riders lucky to be close to a driver. Surge pricing largely avoids this by increasing prices, moving from random allocation towards a price mechanism. While allocative efficiency benefits riders, it has no effect on drivers. The value of getting a trip (relative to being idle) only varies across available drivers with their distance to the rider: all drivers would get the same earnings if they did the trip, but those who are closer would spend less time completing it. Surge pricing has no effect on the matching process, so it does not help reallocate trips towards drivers who are closer.

The value of getting a trip is homogeneous across drivers because of the form I assume for utility. I do not take into account the fact that drivers might have different preferences for trips to certain neighborhoods. Surge pricing, however, is unlikely to help allocate trips more efficiently along this dimension because drivers observe little information before accepting a trip. They do not know the destination; all they know is the surge multiplier and how long it will take to pick up the rider.

Optimal Uniform Pricing

If Uber could only do uniform pricing—because of regulation, for instance—it would reoptimize and set the multiplier at a different level from the average multiplier in the status quo. To determine the level at which Uber would set it, I assume that, to maximize long-run shareholder value, Uber maximizes a weighted sum of current profits, rider surplus, and driver surplus:

$$\max \alpha^{\Pi} \Pi + \alpha^R RS + \alpha^D DS. \quad (16)$$

Uber sets the uniform multiplier that maximizes this objective function.

I find weights that are consistent with how Uber prices in the data. As a normalization, I set $\alpha^R = 1$. I then find values of (α^{Π}, α^D) such that, among the surge pricing policies in Figure 8, the status quo pricing policy maximizes (16). This problem is underdetermined (I back out two weights based on a one-dimensional maximization). However, all nonnegative solutions lead to almost identical results. Uber always gives a small weight to driver surplus and profits, which is as expected, given that the average surge multiplier in the

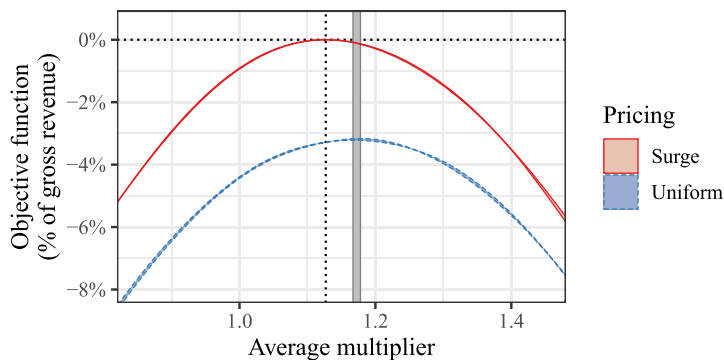


FIGURE 10.—Uber’s objective function. *Note:* This figure shows Uber’s objective function $\Pi + \alpha^R RS + \alpha^D DS$ for the pricing policies in Figure 8. The very narrow shaded areas between lines represent the full range of values that the objective function can take for all nonnegative combinations of weights (α^R, α^D) that rationalize how Uber prices in the data. The vertical and horizontal dotted lines highlight the status quo. The shaded area between the vertical gray lines represents the range of uniform multipliers that maximize Uber’s objective function with uniform pricing.

data is close to the one that maximizes rider surplus: the sum of α^R and α^D is always less than 0.7.

Figure 10 plots Uber’s objective function for all different solutions (α^R, α^D); it is almost identical for all solutions. With surge pricing, welfare always peaks at the status quo, as it should: that is the condition I used to back out the objective function. Under uniform pricing, the optimal multiplier ranges between 1.168 and 1.178, depending on the specific weights. The average is 1.174, which I hereafter take as the baseline value for the optimal uniform multiplier. My main findings remain virtually unchanged if I use other values between 1.168 and 1.178, or if I consider more general objective functions than (16) (see Supplemental Appendix 5.3).

The optimal uniform multiplier (1.174) is higher than the average multiplier in the data. Castillo, Knoepfle, and Weyl (2023) explain why. Driver scarcity is bad for riders because they are matched to far-away drivers. But it is also bad for drivers, who must spend a long time picking up riders, inefficiently using their time right when they are most needed. A negative feedback loop begins: driver scarcity leads to inefficient driver time use, further fueling driver scarcity. Rider surplus, driver surplus, and profit all decrease in a situation that Castillo, Knoepfle, and Weyl call a *wild-goose chase*.

The implication for the market is that at any given time and place, it is worse for Uber to set prices too low—which might trigger a wild-goose chase—than too high. That is not a big issue with surge pricing: the algorithm automatically avoids prices that are too low. But with uniform pricing, it is optimal to set a high multiplier to avoid wild-goose chases and a drop in the objective function. Supplemental Appendix 5.5 provides evidence that that is why the optimal uniform multiplier is higher.

Figure 11 summarizes welfare effects. Relative to surge pricing, the welfare effects of a fixed uniform multiplier are those shown in Figure 9: welfare goes down, mostly because of a drop in rider surplus. With the reoptimized uniform multiplier—the one that maximizes (16)—prices are higher, increasing profits and driver surplus but further decreasing rider surplus. The arrows show the net welfare effects of surge pricing, relative

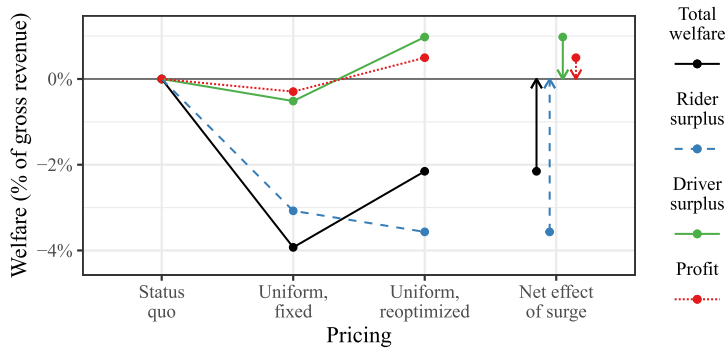


FIGURE 11.—Welfare for different pricing policies. *Note:* This figure plots total welfare, rider surplus, driver surplus, and profit for three counterfactuals: the status quo (surge pricing), a uniform multiplier fixed at 1.174, the average multiplier in the data, and a reoptimized uniform multiplier. Numbers are measured relative to surge pricing. The arrows show the net welfare effect of surge pricing, relative to a reoptimized uniform multiplier.

to the reoptimized uniform multiplier. Rider surplus increases by 3.57% of gross revenue: a more efficient trip allocation and time savings increase rider surplus (lower prices further increase it slightly). In contrast, there is a net decrease in driver surplus and profits of 0.98% and 0.50%, respectively, of gross revenue: the decrease from lower prices outweighs the small increase from time savings. Total welfare goes up by 2.15% of gross revenue.

Uber could also do surge pricing in a manner that does not hurt drivers relative to uniform pricing. This is visualized in the upper right panel in Figure 8: start from a uniform multiplier of 1.174 and move straight left until reaching the line for surge pricing, at an average multiplier of 1.160. Total welfare and rider surplus increase by 3.27% and 3.11% of gross revenue, respectively. Profits go up by 0.07% of gross revenue. Compared to the results presented in Figure 11, this form of surge pricing moderates price decreases to avoid hurting drivers. Therefore, riders benefit less, and total welfare increases more because it gets closer to its maximum at 1.4.

Robustness

Table IV shows that the results in Figure 11 are robust to higher long-run elasticities. Magnitudes differ, but the main qualitative results still hold. This means that the main results are not driven by long-run elasticities. Instead, they are mainly driven by short-run elasticities and by the matching technology. Appendix E shows additional robustness results: if demand is more elastic (which could be the case if public transit is better), if riders have a higher value of time, if riders have rational expectations about pickup times, and if the probability that drivers accept trips varies with multipliers. The main qualitative results hold in all cases.

6.3. Heterogeneity Among Riders and Drivers

Figure 12a depicts rider surplus for different income levels.⁵⁰ Comparing surge pricing relative to a fixed uniform multiplier (i.e., accounting for welfare effects from trip allo-

⁵⁰For occasional riders, the surplus goes down (relative to surge pricing) by \$0.12 per session with a fixed uniform multiplier, and by \$0.28 per session with the reoptimized uniform multiplier.

TABLE IV
WELFARE EFFECTS WITH DIFFERENT LONG-RUN ELASTICITIES.

Market (1)	Demand elasticity (2)	Supply elast. factor (3)	Fixed uniform multiplier				Reoptimized uniform multiplier				
			Total welfare (4)	Rider surplus (5)	Driver surplus (6)	Profit (7)	Mult. (8)	Total welfare (9)	Rider surplus (10)	Driver surplus (11)	Profit (12)
Baseline	-0.633	1	-3.93%	-3.08%	-0.51%	-0.29%	1.174	-2.15%	-3.57%	0.98%	0.50%
Higher supply elasticity	-	1.5	-4.47%	-3.42%	-0.69%	-0.33%	1.238	-1.64%	-4.20%	1.44%	1.12%
	-	2	-4.84%	-3.79%	-0.62%	-0.39%	1.281	-0.79%	-4.60%	2.12%	1.65%
Higher demand elasticity	-1	-	-3.82%	-2.81%	-0.64%	-0.32%	1.177	-2.67%	-3.04%	0.27%	0.16%
	-1.5	-	-3.39%	-2.56%	-0.63%	-0.25%	1.152	-3.29%	-2.63%	-0.50%	-0.08%
Higher supply and demand elasticities	-1	1.5	-4.20%	-3.11%	-0.69%	-0.35%	1.220	-2.08%	-3.50%	0.77%	0.68%
	-1.5	2	-4.49%	-3.16%	-0.96%	-0.32%	1.237	-2.99%	-3.51%	0.01%	0.53%

Note: This table shows how welfare changes with higher long-run elasticities. The first row presents results with the baseline elasticities; all other rows present results with alternative elasticities. Columns (2) and (3) describe how elasticities are modified relative to the baseline. A dash means that parameters are the same as in the baseline market. Columns (4)–(7) show welfare with a uniform multiplier set at the average multiplier in the data. Columns (9)–(11) show welfare with a uniform multiplier set at the level that maximizes Uber’s objective function, which I show in column (8).

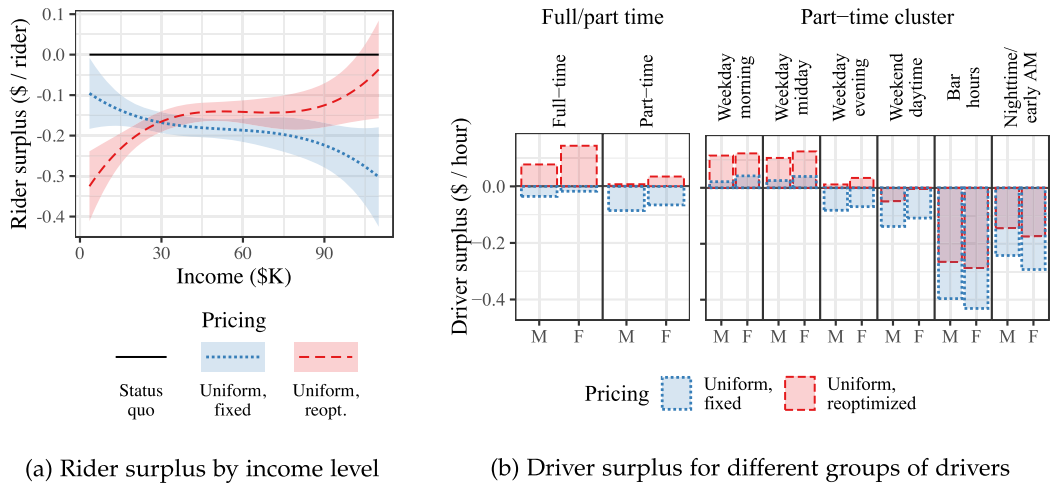


FIGURE 12.—Heterogeneity in welfare effects within riders and drivers. *Note:* Subfigure (a) shows rider surplus by income level for different pricing policies. For each policy, I fit a cubic spline with five degrees of freedom based on simulated data for 50 weeks. The lines are differences between these fits. Shaded areas are 95% confidence intervals. Subfigure (b) shows the surplus of different types of drivers for uniform pricing counterfactuals, relative to surge pricing.

cation and time savings), surge pricing benefits riders at all income levels. High-income riders benefit most: they have a higher value of time and lose more when they do not get a trip. Comparing surge pricing relative to the reoptimized uniform multiplier (also accounting for lower overall prices), riders at all income levels benefit, but low-income riders benefit most: they are price sensitive, and thus prefer low prices. High-income riders, in contrast, prefer higher prices that skim other riders and bring in more drivers, resulting in more reliable trips with lower pickup times.

Figure 12b shows driver surplus with uniform pricing for different types of drivers, relative to surge pricing. The heterogeneity in this figure is mainly driven by the fact that certain types of drivers are more likely to work when surge pricing is high (Section 5.2.3).⁵¹ Full-time drivers are hurt the most by surge pricing: for every hour of work, the surplus of full-time women goes down by \$0.13, and the surplus of full-time men decreases by \$0.08. This is so because they tend not to work during bar hours, when surge multipliers are highest (see Figure 2). Full-time men partially compensate by working during the early morning, when surge multipliers are also high, but full-time women do not—hence, they are hurt the most.

On average, part-time drivers are close to indifferent between surge and uniform pricing. However, there is substantial heterogeneity depending on the times they tend to work. Whereas drivers who work during times when surge is highest (bar hours and the early morning) benefit from surge pricing by \$0.15–0.3 per hour they work, drivers who work during the morning and midday on weekdays (when multipliers are lowest) are worse off by around \$0.1 per hour. Supplemental Appendix 5.2 presents further results about the heterogeneity of the welfare effects on drivers.

⁵¹Driver types also differ in terms of how responsive their entry and exit decisions are to expected earnings (Table III). However, that only has a minor (second-order) effect on driver surplus since it only affects marginal drivers who are close to indifferent between working and not working.

7. CONCLUSION

In the debate about the desirability of surge pricing, standard arguments about efficiency gains are challenged by concerns that individual market participants might be hurt. I provide evidence that supports both sides of the debate. I find efficiency gains that lead to higher welfare. I also find that riders benefit substantially from surge pricing across all income levels. Riders' frequent complaints might arise because they are not aware that, without surge pricing, they would have to wait longer for less reliable trips. On the other hand, my findings about drivers' surplus—an overall decrease with winners and losers depending on the times of the week they work—suggest that drivers might be right to complain about surge pricing.

My findings are quite different from those in previous works that conclude that surge pricing benefits all sides of the market (Cachon, Daniels, and Lobel (2017), Ming et al. (2023)). The key difference is that I microfound and estimate matching frictions (and how agents are affected by them) using high-resolution data. Thus, I can capture important features of matching frictions that drive my main results: these frictions hurt riders more than drivers because of the value of time and because of trip misallocation, and they lead to the failures predicted by Castillo, Knoepfle, and Weyl (2023). This highlights that understanding markets in which search and matching frictions are central requires modeling those frictions carefully (Gavazza and Lizzeri (2021)).

APPENDIX A: RESULTS ABOUT EQUILIBRIUM

PROPOSITION 3—Existence: *A fixed point $\mathbf{x}^P \in \mathcal{X}$ of $f^P(\cdot)$ exists.*

PROOF: $f^P(\cdot)$ is continuous: all functions that are involved are continuous. $\mathcal{X}$ is bounded by the lowest and highest utilities agents can get. $\mathcal{X}$ is, thus, a convex, compact space, so $f^P(\cdot)$ has a fixed point by Brouwer's fixed point theorem. *Q.E.D.*

ASSUMPTION 3: There exists some $\delta < 1$ such that $(f^P(\mathbf{x}) - f^P(\mathbf{x}')) \cdot (\mathbf{x} - \mathbf{x}') < \delta \|\mathbf{x} - \mathbf{x}'\|^2$ for every $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$.

I cannot prove this assumption, but it holds for every pair $(\mathbf{x}, \mathbf{x}')$ I have tried. It is justified by a simple intuition. As beliefs change from $\mathbf{x}$ to $\mathbf{x}'$, drivers are more likely to work in locations and times with higher earnings, and more riders will request trips at times and locations with higher utilities. Those times and locations get congested and result in lower utilities. As long as these congestion effects are stronger than complementarities between riders and drivers, there will be a negative correlation between changes in beliefs and changes in empirical averages, $(f^P(\mathbf{x}) - f^P(\mathbf{x}')) \cdot (\mathbf{x} - \mathbf{x}') < 0$, which is a stronger condition than Assumption 3.

PROPOSITION 4—Uniqueness: *Under Assumption 3, $f^P(\cdot)$ has a unique fixed point.*

PROOF: Let $g_\gamma : \mathcal{X} \rightarrow \mathcal{X}$ be defined by $g_\gamma(\mathbf{x}) = (1 - \gamma)\mathbf{x} + \gamma f^P(\mathbf{x})$, where $\gamma \in (0, 1)$. The set of fixed points of f^P and g_γ is the same. I will now show that there exists some γ such that g_γ is a contraction mapping. This implies, by the contraction mapping theorem, that g_γ has a unique fixed point—and, hence, so does f^P .

$f^P(\cdot)$ is uniformly continuous because every function that is involved is uniformly continuous. Thus, there exists $\beta < \infty$ such that $\frac{\|f^P(\mathbf{x}) - f^P(\mathbf{x}')\|}{\|\mathbf{x} - \mathbf{x}'\|} < \beta \forall \mathbf{x}, \mathbf{x}' \in \mathcal{X}$. For all $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$,

$\|g_\gamma(\mathbf{x}) - g_\gamma(\mathbf{x}')\|^2 = (1 - \gamma)^2 \|\mathbf{x} - \mathbf{x}'\|^2 + 2\gamma(1 - \gamma)(f^P(\mathbf{x}) - f^P(\mathbf{x}')) \cdot (\mathbf{x} - \mathbf{x}') + \gamma^2 \|f^P(\mathbf{x}) - f^P(\mathbf{x}')\|^2 < [(1 - \gamma)^2 + 2\gamma(1 - \gamma)\delta + \gamma^2\beta^2] \|\mathbf{x} - \mathbf{x}'\|^2$. Since β is bounded, the term in brackets is $1 + 2(\delta - 1)\gamma + O(\gamma^2)$, which is less than 1 for small-enough $\gamma > 0$. Thus, there exist some $\gamma > 0$ and some $\eta \in (0, 1)$ such that $\|g_\gamma(\mathbf{x}) - g_\gamma(\mathbf{x}')\| \leq \eta \|\mathbf{x} - \mathbf{x}'\|$ for all $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$. Q.E.D.

ASSUMPTION 4: Simulation draws take the form $\hat{f}^P(\mathbf{x}) = f^P(\mathbf{x}) + \varepsilon$, where ε is a vector of mean-zero, bounded-variance random variables that are independent across simulations.

PROPOSITION 5—Convergence: Let $(\mathbf{x}_n)_{n=0}^\infty$ be a sequence over $\mathcal{X}$ defined iteratively by equation (10), where $\mathbf{x}_0 \in \mathcal{X}$. Under Assumptions 3 and 4, $\mathbf{x}_n \xrightarrow{P} \mathbf{x}^P$.

PROOF: Let $V_n = \mathbb{E}[\|\mathbf{x}_n - \mathbf{x}^P\|^2]$. By the independence of ε , $V_n = \mathbb{E}[\|(1 - \gamma_n)\mathbf{x}_{n-1} + \gamma_n f_n^P(\mathbf{x}_{n-1}) - \mathbf{x}^P\|^2] + \gamma_n^2 \mathbb{E}[\|\varepsilon\|^2]$. The algebra in the proof of Proposition 4 implies that $\|\mathbf{x}_n - \mathbf{x}^P\|^2 < [(1 - \gamma_n)^2 + 2\gamma_n(1 - \gamma_n)\delta + \gamma_n^2\beta^2] \|\mathbf{x}_{n-1} - \mathbf{x}^P\|^2$ pointwise, so $V_n < [(1 - \gamma_n)^2 + 2\gamma_n(1 - \gamma_n)\delta + \gamma_n^2\beta^2] V_{n-1} + \gamma_n^2 \Gamma$, where Γ is some finite real number such that $\mathbb{E}[\|\varepsilon\|^2] < \Gamma$. This recursion over V_n converges to zero (see Broadie, Cicek, and Zeevi (2011), online appendix), so $\mathbf{x}_n$ converges in probability to $\mathbf{x}^P$. Q.E.D.

APPENDIX B: PROOFS OF PROPOSITIONS 1 AND 2

PROOF OF PROPOSITION 1: We can define $\eta_i = \epsilon_{it} - \mathbb{E}[\epsilon_i | \tilde{\mathbf{m}}_t, x_i, w_{it}^0; l, h]$ to obtain equation (11). Since (a) p_i is a deterministic function of $(\tilde{\mathbf{m}}_t, \bar{p}_i, l)$, (b) $\bar{p}_i$ is part of x_i , and (c) $w_i = w_{it}^0 + \xi_i$, then $\mathbb{E}[\eta_i | p_i, w_i, \tilde{\mathbf{m}}_t, x_i, w_{it}^0, l, h] = \mathbb{E}[\eta_i | w_i, \tilde{\mathbf{m}}_t, x_i, l, h, \xi_i] = \mathbb{E}[\epsilon_i | w_i, \tilde{\mathbf{m}}_t, x_i, l, h, \xi_i] - \mathbb{E}[\mathbb{E}[\epsilon_i | \tilde{\mathbf{m}}_t, x_i, w_{it}^0; l, h] | w_i, \tilde{\mathbf{m}}_t, x_i, l, h, \xi_i]$. By assumption, $\xi_i \perp (\epsilon_i, x_i, \tilde{\mathbf{m}}_t)$, and by definition, $\xi_i \perp (w_{it}^0, l, t)$. The dependence on ξ_i can thus be dropped out of both terms in the last expression, which is then zero. Thus, $\mathbb{E}[\eta_i | p_i, w_i, \tilde{\mathbf{m}}_t, x_i, w_{it}^0, l, h] = 0$.

We can parameterize the model (ignoring indices, to simplify notation) as $U(z; \theta) + \eta$, where $U(z; \theta) = \alpha(x, l, h) + \beta(x)p(z) + \gamma(x)w + g(z)$, $z = (\tilde{\mathbf{m}}, w, w^0, x, l, t)$, $\theta = (\alpha, \beta, \gamma, g) \in \Theta$, and $\Theta = \mathcal{A} \times \mathcal{B} \times \Gamma \times \mathcal{G}$ is the cartesian product of all functions $\alpha(x, l, h)$, $\beta(x)$, $\gamma(x)$, and all functions $g(z)$ that are Lipschitz-continuous in $\tilde{\mathbf{m}}$. Take $\theta = (\alpha, \beta, \gamma, g) \in \Theta$ and $\theta' = (\alpha', \beta', \gamma', g') \in \Theta$ such that $\gamma(x) \neq \gamma'(x)$ with positive probability. Since the only term in $U(z; \theta)$ that varies with w is $\gamma(x)w$, and w is not constant, then $U(z; \theta) \neq U(z; \theta')$ with positive probability. Since η has full support (because ϵ has full support), then $\Pr(U(z; \theta) + \eta > 0 | z) \neq \Pr(U(z; \theta') + \eta > 0 | z)$ with positive probability. Thus, γ is identified.

Now take $\theta = (\alpha, \beta, \gamma, \phi) \in \Theta$ and $\theta' = (\alpha', \beta', \gamma', \phi') \in \Theta$ such that $\beta(x) \neq \beta'(x)$ with positive probability. $p(z)$ has nonremovable discontinuities in $\tilde{\mathbf{m}}$ for all x , which means that $(\beta(x) - \beta'(x))p(z)$ has nonremovable discontinuities with positive probability. Those discontinuities cannot be offset by the other three terms in $U(z; \theta)$ because $\alpha(x, l, h)$ and $\gamma(x)w$ are constant in $\tilde{\mathbf{m}}$ and $g(z)$ is Lipschitz-continuous in $\tilde{\mathbf{m}}$. Thus, $U(z; \theta) - U(z; \theta')$ has nonremovable discontinuities with positive probability and, therefore, $U(z; \theta) \neq U(z; \theta')$ with positive probability. This, coupled with the fact that η has full support, implies that $\Pr(U(z; \theta) + \eta > 0 | z) \neq \Pr(U(z; \theta') + \eta > 0 | z)$ with positive probability. Thus, β is identified. Q.E.D.

PROOF OF PROPOSITION 2: We can write $\zeta_{j,t+1}^k = g^M(\tilde{\mathbf{m}}_{tk}; l, h, d) + \psi_{j,t+1}^k$, where $\mathbb{E}[\psi_{j,t+1}^k | \tilde{\mathbf{m}}_{tk}; l, h, d] = 0$ by the definition of conditional mean. This yields $y_{j,t+1}^k = \omega_{lkhd} + \delta_d v_{dkt} + g^M(\tilde{\mathbf{m}}_{tk}; l, h, d) + \psi_{j,t+1}^k$. Since $\psi_{j,t+1}^k$ are i.i.d. Gumbel random variables, the movement probabilities are given by equation (12). And since v_{dkt} is a deterministic function of $\tilde{\mathbf{m}}_{tk}$, $\mathbb{E}[\psi_{j,t+1}^k | v_{dkt}, \tilde{\mathbf{m}}_{tk}, l, h, d] = \mathbb{E}[\psi_{j,t+1}^k | \tilde{\mathbf{m}}_{tk}, l, h, d] = 0$.

We can parameterize the model (ignoring indices j, t , and d) as $y^k(z; \theta) + \psi^k$, where $y^k(z; \theta) = \omega_k(z) + \delta v_k(z) + g_k^M(z)$, $z = (\mathbf{m}, l, h, d)$, $\theta = (\omega, \delta, g) \in \Theta$, and $\Theta = \Omega \times \mathcal{D} \times \mathcal{G}^M$ is the cartesian product of all functions $\omega(z) = (\omega_k(z))_k$, all vectors $\delta = (\delta_d)_d$, and all functions $g^M(z) = (g_k^M(z))_k$ that are Lipschitz-continuous in $\tilde{\mathbf{m}}$. Take $\theta = (\delta, \omega, g^M) \in \Theta$ and $\theta' = (\delta', \omega', g'^M) \in \Theta$ such that $\delta \neq \delta'$. The term $\delta v(z)$ has at least one discontinuity in $\tilde{\mathbf{m}}$. $g^M(z)$ is Lipschitz-continuous in $\tilde{\mathbf{m}}$, and no other term depends on $\tilde{\mathbf{m}}$. Thus, $y^k(z; \theta) \neq y^k(z; \theta')$ for some k with positive probability. Since ψ^k are i.i.d. and have full support, the mapping from y to movement choice probabilities is injective (see [Berry, Gandhi, and Haile \(2013\)](#)), which implies that, for some k , $\Pr(k = \arg\max_{k'} \{y^{k'}(z; \theta) + \eta^{k'}\} | z) \neq \Pr(k = \arg\max_{k'} \{y^{k'}(z; \theta') + \eta^{k'}\} | z)$ with positive probability. Thus, δ is identified. *Q.E.D.*

APPENDIX C: COMPUTATION OF THE VALUE OF STAYING

Let $q_\Delta(\mathbf{s}_t)$ be the match probability in Δ periods, conditional on $\mathbf{s}_t$. Let $\pi_\Delta^M(\mathbf{s}_t)$, $T_\Delta(\mathbf{s}_t)$, and $T_\Delta^2(\mathbf{s}_t)$ be, respectively, the expected earnings, duration, and duration squared until the end of the next trip if the trip starts in Δ periods, given $\mathbf{s}_t$. Also, let $y_\Delta(\mathbf{s}_t) = (q_\Delta(\mathbf{s}_t), \pi_\Delta^M(\mathbf{s}_t), T_\Delta(\mathbf{s}_t), T_\Delta^2(\mathbf{s}_t))$. Given the functional form of the opportunity cost, the utility until the end of the trip can be written as⁵²

$$v_{d,\Delta}^M(\tilde{\mathbf{s}}_{jt}) = v_{d,\Delta}^M(\tau_{jt}, y_\Delta(\mathbf{s}_t)) = \pi_\Delta^M(\mathbf{s}_t) - \left(\frac{1}{2} + \Delta + \tau_{jt}\right) \kappa_d T_\Delta(\mathbf{s}_t) - \frac{\kappa_d}{2} T_\Delta^2(\mathbf{s}_t). \quad (17)$$

Starting from $\{y_\Delta(\mathbf{s}_t)\}_\Delta$ and τ_{jt} , one can use this expression to compute $v_{d,\Delta}^M(\tilde{\mathbf{s}}_{jt})$, which one can then use to compute the value of staying $v_{d,0}^S(\tilde{\mathbf{s}}_{jt})$ using equation (6).

To estimate $y_\Delta(\mathbf{s}_t)$, I run models of the form (9) with the functional form described above, with the appropriate left-hand-side variable. To estimate $T_\Delta(\mathbf{s}_t)$ for $\Delta = 4$, for instance, I take observations in which a driver was available and was matched 4 periods later. I regress the realized trip duration on the state.

APPENDIX D: DESCRIPTIVE EVIDENCE ABOUT DRIVER EXIT

Table V presents regressions of dummies for whether drivers stop working on multipliers. The first three regressions show negative and significant coefficients. An increase in the surge multiplier from 1 to 1.5 reduces the probability of exiting by around 23%. The last three regressions control for unrounded multipliers in order to use only variation from rounding, as in Section 4.2. The coefficients are similar to those in the first three columns, although they are not statistically significant.

APPENDIX E: ROBUSTNESS OF WELFARE EFFECTS

Like Table IV, Table VI presents results of rerunning my main counterfactuals to compute welfare effects under different model assumptions.

⁵²This expression is valid if $\Delta \geq \tau - 1$. $v_{d,\Delta}^M(\tilde{\mathbf{s}}_{jt})$ is higher if $\Delta < \tau - 1$, but this case is only relevant during a small fraction of the periods in which drivers leave.

TABLE V
RESPONSE OF SHIFT LENGTH TO MULTIPLIERS.

	<i>Dependent variable: Driver stopped working</i>					
	(1)	(2)	(3)	(4)	(5)	(6)
Multiplier	-0.0101 (0.0016)			-0.0113 (0.0100)		
3-ring avg. mult.		-0.0104 (0.0016)			-0.0111 (0.0134)	
5-ring avg. mult.			-0.0107 (0.0017)			-0.0086 (0.0131)
Loc. × week-hour	✓	✓	✓	✓	✓	✓
Rounding controls				✓	✓	✓
Observations	1,580,853	1,580,824	1,580,824	1,580,853	1,580,824	1,580,824

Note: This table presents results of regressions of a dummy for whether riders decided to stop working. There is one observation for every period that a driver was available. The average dependent variable is 0.0216. In columns (1) and (4), the regressor is the multiplier in the driver's location. In columns (2), (3), (5), and (6), the regressor is the average multiplier among the locations in the 3 and 5 rings that are nearest to the driver. All regressions include location-by-hour-of-the-week fixed effects. Columns (4)–(7) control for the unrounded multipliers in the relevant locations.

TABLE VI
WELFARE EFFECTS WITH ALTERNATIVE MODEL PARAMETERS.

Market (1)	Pickup factor (2)	Price factor (3)	Demand elasticity (4)	Fixed uniform multiplier			Reoptimized uniform multiplier					
				Total welfare (5)	Rider surplus (6)	Driver surplus (7)	Profit (8)	Mult. (9)	Total welfare (10)	Rider surplus (11)	Driver surplus (12)	Profit (13)
Baseline	1	1	-0.633	-3.93%	-3.08%	-0.51%	-0.29%	1.174	-2.15%	-3.57%	0.98%	0.50%
More elastic demand	1.50	1.5	-1.0	-3.41%	-2.53%	-0.55%	-0.33%	1.181	-1.94%	-3.18%	0.81%	0.45%
	2.00	2.0	-1.5	-2.81%	-2.13%	-0.40%	-0.26%	1.191	-1.66%	-2.83%	0.72%	0.44%
Lower pickup time coefficient	0.75	-	-	-3.91%	-3.22%	-0.45%	-0.22%	1.168	-3.09%	-3.78%	0.47%	0.41%
	0.50	-	-	-4.01%	-3.32%	-0.46%	-0.22%	1.157	-3.57%	-3.91%	0.15%	0.19%
Higher price coefficient	-	1.5	-	-4.95%	-4.03%	-0.66%	-0.24%	1.192	-4.23%	-5.34%	0.68%	0.41%
	-	2.0	-	-5.23%	-4.31%	-0.67%	-0.23%	1.212	-4.72%	-5.24%	0.19%	0.33%
Higher price coef., lower price coef.	0.75	1.5	-	-4.23%	-3.56%	-0.42%	-0.23%	1.172	-3.63%	-4.54%	0.66%	0.26%
	0.50	2.0	-	-4.33%	-3.70%	-0.32%	-0.28%	1.162	-4.53%	-5.05%	0.36%	0.17%
Expected pickup time	-	-	-	-3.76%	-2.98%	-0.53%	-0.23%	1.172	-2.51%	-3.24%	0.34%	0.37%
Endogenous acceptance	-	-	-	-3.91%	-3.14%	-0.60%	-0.16%	1.168	-2.58%	-3.37%	0.35%	0.43%

Note: This table shows welfare, as in Figure 11, with certain adjustments to the model. The first row presents the baseline model. Columns (2)–(4) specify how model parameters are modified; dashes indicate that parameters are the same as in the baseline market. In “More elastic demand,” all demand elasticities are higher (both short- and long-run); in the trip request model, the pickup time coefficient is lower by the factor in column (2) and the price coefficient is lower by the factor in column (3), and the long-run demand elasticity is the one specified in column (4). In “Lower pickup time coef.,” the pickup time coefficient in the demand model is lower by the factor in column (2). In “Higher price coefficient,” the price coefficient in the demand model is higher by the factor in column (3). “Higher price, lower pickup coef.” combines the previous two adjustments. In “Exp. pickup time,” riders base trip request decisions on the expected pickup time conditional on the displayed pickup time (which I estimate with a cubic fit). In “Endo. acceptance,” drivers’ trip acceptance responds to surge multipliers according to the flexible fit shown in Supplemental Appendix 4.1. Columns (5)–(8) show welfare with a uniform multiplier in the data. Columns (10)–(13) show welfare with the uniform multiplier in column (9), which maximizes Uber’s objective function.

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The replication package for this paper is available at <https://doi.org/10.5281/zenodo.15191154>. The authors were granted an exemption to publish parts of their data because either access to these data is restricted or the authors do not have the right to republish them. However, the authors included in the package, on top of the codes and the parts of the data that are not subject to the exemption, a simulated or synthetic dataset that allows running the codes. The Journal checked the data and the codes for their ability to generate all tables and figures in the paper and approved online appendices. Whenever the available data allowed, the Journal also checked for their ability to reproduce the results. However, the synthetic/simulated data are not designed to produce the same results. Given the highly demanding nature of the algorithms, the reproducibility checks were run on a simplified version of the code, which is also available in the replication package.